WHO also encourages international and national funders to support relevant learning opportunities.

5. Case management

Background
Malaria case management, consisting of early diagnosis and prompt effective treatment, remains a vital component of malaria control and elimination strategies. The WHO Guidelines for the treatment of malaria were first developed in 2006 and have been revised periodically, with the most recent edition published in 2015. WHO guidelines contain recommendations on clinical practice or public health policy intended to guide end-users as to the individual or collective actions that can or should be taken in specific situations to achieve the best possible health outcomes. Such recommendations are also designed to help the user to select and prioritize interventions from a range of potential alternatives. The third edition of the WHO Guidelines for the treatment of malaria consolidated here contains updated recommendations based on new evidence particularly related to dosing in children, and also includes recommendations on the use of drugs to prevent malaria in groups at high risk.

Since publication of the first edition of the *Guidelines for the treatment of malaria* in 2006 and the second edition in 2010, all countries in which *P. falciparum* malaria is endemic have progressively updated their treatment policy from use of monotherapy with drugs such as chloroquine, amodiaquine and sulfadoxine–pyrimethamine (SP) to the currently recommended artemisinin-based combination therapies (ACT). The ACTs are generally highly effective and well tolerated. This has contributed substantially to reductions in global morbidity and mortality from malaria. Unfortunately, resistance to artemisinins has arisen recently in *P. falciparum* in South-East Asia, which threatens these gains.

Core principles
The following core principles were used by the Guidelines Development Group that drew up the Guidelines for the Treatment of Malaria.

1. Early diagnosis and prompt, effective treatment of malaria
Uncomplicated falciparum malaria can progress rapidly to severe forms of the disease, especially in people with no or low immunity, and severe falciparum malaria is almost always fatal without treatment. Therefore, programmes should ensure access to early diagnosis and prompt, effective treatment within 24–48 h of the onset of malaria symptoms.

2. Rational use of antimalarial agents
To reduce the spread of drug resistance, limit unnecessary use of antimalarial drugs and better identify other febrile illnesses in the context of changing malaria epidemiology, antimalarial medicines should be administered only to patients who truly have malaria. Adherence to a full treatment course must be promoted. Universal access to parasitological diagnosis of malaria is now possible with the use of quality-assured rapid diagnostic tests (RDTs), which are also appropriate for use in primary health care and community settings.

3. Combination therapy
Preventing or delaying resistance is essential for the success of both national and global strategies for control and eventual elimination of malaria. To help protect current and future antimalarial medicines, all episodes of malaria should be treated with at least two effective antimalarial medicines with different mechanisms of action (combination therapy).

4. Appropriate weight-based dosing
To prolong their useful therapeutic life and ensure that all patients have an equal chance of being cured, the quality of antimalarial drugs must be ensured, and antimalarial drugs must be given at optimal dosages. Treatment should maximize the likelihood of rapid clinical and parasitological cure and minimize transmission from the treated infection. To achieve this, dosage regimens should be based on the patient’s weight and should provide effective concentrations of antimalarial drugs for a sufficient time to eliminate the infection in all target populations.

Please refer to *Malaria case management: operations manual* *[197]*.

5.1 Diagnosing malaria

Suspected malaria
The signs and symptoms of malaria are non-specific. Malaria is suspected clinically primarily on the basis of fever or a history of fever. There is no combination of signs or symptoms that reliably distinguishes malaria from other causes of fever; diagnosis based only on clinical features has very low specificity and results in overtreatment. Other possible causes of fever and whether alternative or additional treatment is required must always be carefully considered. The focus of malaria diagnosis should be to identify patients who truly have malaria, to guide rational use of antimalarial medicines.

In malaria-endemic areas, malaria should be suspected in any patient presenting with a history of fever or temperature ≥ 37.5 °C and no other obvious cause. In areas in which malaria transmission is stable (or during the high-transmission period of seasonal malaria), malaria should also be suspected in children with palmar pallor or a haemoglobin concentration of < 8 g/dL. High-transmission settings include many parts of sub-Saharan Africa and some parts of Oceania.

In settings where the incidence of malaria is very low, parasitological diagnosis of all cases of fever may result in considerable expenditure to detect only a few patients with malaria. In these settings, health workers should be trained to identify patients who may have been exposed to malaria (e.g. recent travel to a malaria-endemic area without protective measures) and have fever or a history of fever with no other obvious cause, before they conduct a parasitological test.

In all settings, suspected malaria should be confirmed with a parasitological test. The results of parasitological diagnosis should be available within a short time (< 2 h) of the patient presenting. In settings where parasitological diagnosis is not possible, a decision to provide antimalarial treatment must be based on the probability that the illness is malaria.

In children < 5 years, the practical algorithms for management of the sick child provided by the WHO–United Nations Children's Fund (UNICEF) strategy for Integrated Management of Childhood Illness *[198]* should be used to ensure full assessment and appropriate case management at first-level health facilities and at the community level.

Parasitological diagnosis
The benefit of parasitological diagnosis relies entirely on an appropriate management response of health care providers. The two methods used routinely for parasitological diagnosis of malaria are light microscopy and immunochromatographic RDTs. The latter detect parasite-specific antigens or enzymes that are either genus or species specific.

Both microscopy and RDTs must be supported by a quality assurance programme. Antimalarial treatment should be limited to cases with positive tests, and patients with negative results should be reassessed for other common causes of fever and treated appropriately.

In nearly all cases of symptomatic malaria, examination of thick and thin blood films by a competent microscopist will reveal malaria parasites. Malaria RDTs should be used if quality-assured malaria microscopy is not readily available. RDTs for detecting PfHRP2 can be useful for patients who have received incomplete antimalarial treatment, in whom blood films can be negative. This is particularly likely if the patient received a recent dose of an artemisinin derivative. If the initial blood film examination is negative in patients with manifestations compatible with severe malaria, a series of blood films should be examined at 6–12 h intervals, or an RDT (preferably one detecting PfHRP2) should be performed. If both the slide examination and the RDT results are negative, malaria is extremely unlikely, and other causes of the illness should be sought and treated.

This document does not include recommendations for use of specific RDTs or for interpreting test results. For guidance, see the WHO manual *Universal access to malaria diagnostic testing* *[199]*.

Diagnosis of malaria
In patients with suspected severe malaria and in other high-risk groups, such as patients living with HIV/AIDS, absence or delay of parasitological diagnosis should not delay an immediate start of antimalarial treatment.

At present, molecular diagnostic tools based on nucleic-acid amplification techniques (e.g. loop-mediated isothermal amplification or polymerase chain reaction [PCR]) do not have a role in the clinical management of malaria.

Where *P. vivax* malaria is common and microscopy is not available, it is recommended that a combination RDT be used that allows detection of P. vivax (pLDH antigen from *P. vivax*) or pan-malarial antigens (Pan-pLDH or aldolase).

Light microscopy
Microscopy not only provides a highly sensitive, specific diagnosis of malaria when performed well but also allows quantification of malaria parasites and identification of the infecting species. Light microscopy involves relatively high costs for training and supervision, and the accuracy of diagnosis is strongly dependent on the competence of the microscopist. Microscopy technicians may also contribute to the diagnosis of non-malarial diseases.

Although nucleic acid amplification-based tests are more sensitive, light microscopy is still considered the "field standard" against which the sensitivity and specificity of other methods must be assessed. A skilled microscopist can detect asexual parasites at a density of < 10 per µL of blood, but under typical field conditions, the limit of sensitivity is approximately 100 parasites per µL *[200]*. This limit of detection approximates the lower end of the pyrogenic density range. Thus, microscopy provides good specificity for diagnosing malaria as the cause of a presenting febrile illness. More sensitive methods allow detection of an increasing proportion of cases of incidental parasitaemia in endemic areas, thus reducing the specificity of a positive test. Light microscopy has other important advantages:

- low direct costs, if laboratory infrastructure to maintain the service is available;

- high sensitivity, if the performance of microscopy is high;
- differentiation of Plasmodia species;
- determination of parasite densities – notably identification of hyperparasitaemia;
- detection of gametocytaemia;
- allows monitoring of responses to therapy and
- can be used to diagnose many other conditions.

Good performance of microscopy can be difficult to maintain, because of the requirements for adequate training and supervision of laboratory staff to ensure competence in malaria diagnosis, electricity, good quality slides and stains, provision and maintenance of good microscopes and maintenance of quality assurance *[201]* and control of laboratory services.

Numerous attempts have been made to improve malaria microscopy, but none has proven to be superior to the classical method of Giemsa staining and oil-immersion microscopy for performance in typical health care settings *[202]*.

Rapid diagnostic tests
Rapid diagnostic tests (RDTs) are immuno-chromatographic tests for detecting parasite-specific antigens in a finger-prick blood sample. Some tests allow detection of only one species (*P. falciparum*); others allow detection of one or more of the other species of human malaria parasites (*P. vivax*, *P. malariae* and *P. ovale*) *[203][204][205]*. They are available commercially in various formats, e.g. dipsticks, cassettes and cards. Cassettes and cards are easier to use in difficult conditions outside health facilities. RDTs are relatively simple to perform and to interpret, and they do not require electricity or special equipment *[206]*.

Since 2012, WHO has recommended that RDTs should be selected in accordance with the following criteria, based on the results of the assessments of the *WHO Malaria RDT Product Testing programme* *[207]*:

- For detection of *P. falciparum* in all transmission settings, the panel detection score against *P. falciparum* samples should be at least 75% at 200 parasites/µL.
- For detection of *P. vivax* in all transmission settings the panel detection score against *P. vivax* samples should be at least 75% at 200 parasites/µL.
- The false positive rate should be less than 10%.
- The invalid rate should be less than 5%.

Current tests are based on the detection of histidine-rich protein 2 (HRP2), which is specific for *P. falciparum*, pan-specific or species-specific *Plasmodium* lactate dehydrogenase (*p*LDH) or pan-specific aldolase. The different characteristics of these antigens may affect their suitability for use in different situations, and these should be taken into account in programmes for RDT implementation. The tests have many potential advantages, including:

- rapid provision of results and extension of diagnostic services to the lowest-level health facilities and communities;
- fewer requirements for training and skilled personnel (for instance, a general health worker can be trained in 1 day); and
- reinforcement of patient confidence in the diagnosis and in the health service in general.

They also have potential disadvantages, including:

- inability, in the case of *Pf*HRP2-based RDTs, to distinguish new infections from recently and effectively treated infections, due to the persistence of *Pf*HRP2 in the blood for 1–5 weeks after effective treatment;
- the presence in countries in the Amazon region of variable frequencies of HRP2 deletions in *P. falciparum* parasites, making HRP2-based tests not suitable in this region *[208]*;
- poor sensitivity for detecting *P. malariae* and *P. ovale*; and
- the heterogeneous quality of commercially available products and the existence of lot-to-lot variation.

In a systematic review *[209]*, the sensitivity and specificity of RDTs in detecting *P. falciparum* in blood samples from patients in endemic areas attending ambulatory health facilities with symptoms suggestive of malaria were compared with the sensitivity and specificity of microscopy or polymerase chain reaction. The average sensitivity of *Pf*HRP2-detecting RDTs was 95.0% (95% confidence interval [CI], 93.5–96.2%), and the specificity was 95.2% (93.4–99.4%). RDTs for detecting *p*LDH from *P. falciparum* are generally less sensitive and more specific than those for detecting HRP2, with an average sensitivity (95% CI) of 93.2% (88.0–96.2%) and a specificity of 98.5% (96.7–99.4%). Several studies have shown that health workers, volunteers and private sector providers can, with adequate training and supervision, use RDTs correctly and provide accurate malaria diagnoses. The criteria for selecting and procuring RDTs can be found on the WHO website.

Diagnosis with either microscopy or RDTs is expected to reduce overuse of antimalarial medicines by ensuring that treatment is given only to patients with confirmed malaria infection, as opposed to treating all patients with fever *[210]*. Although providers of care may be willing to perform diagnostic tests, they do not, however, always respond appropriately to the results. This is especially true when they are negative. It is therefore important to ensure the accuracy of parasite- based diagnosis and also to demonstrate this to users and to provide them with the resources to manage both positive and negative results adequately *[199]*.

Immunodiagnosis and nucleic acid amplification test methods
Detection of antibodies to parasites, which may be useful for epidemiological studies, is neither sensitive nor specific enough to be of use in the management of patients suspected of having malaria *[211]*.

Techniques to detect parasite nucleic acid, e.g. polymerase chain reaction and loop-mediated isothermal amplification, are highly sensitive and very useful for detecting mixed infections, in particular at low parasite densities that are not detectable by conventional microscopy or with RDTs. They are also useful for studies of drug resistance and other specialized epidemiological investigations *[212]*; however, they are not generally available for large-scale field use in malaria- endemic areas, nor are they appropriate for routine diagnosis in endemic areas where a large proportion of the population may have low-density parasitaemia.

These techniques may be useful for population surveys and focus investigation in malaria elimination programmes.

At present, nucleic acid-based amplification techniques have no role in the clinical management of malaria or in routine surveillance systems *[213]*.

Practice Statement

Diagnosing malaria (2015)

All cases of suspected malaria should have a parasitological test (microscopy or RDT) to confirm the diagnosis.

Both microscopy and RDTs should be supported by a quality assurance programme.

Justification

Prompt, accurate diagnosis of malaria is part of effective disease management. All patients with suspected malaria should be treated on the basis of a confirmed diagnosis by microscopy examination or RDT testing of a blood sample. Correct diagnosis in malaria-endemic areas is particularly important for the most vulnerable population groups, such as young children and non-immune populations, in whom falciparum malaria can be rapidly fatal. High specificity will reduce unnecessary treatment with antimalarial drugs and improve the diagnosis of other febrile illnesses in all settings.

WHO strongly advocates a policy of “test, treat and track” to improve the quality of care and surveillance.

5.2 Treating malaria

5.2.1 Treating uncomplicated malaria

Definition of uncomplicated malaria
A patient who presents with symptoms of malaria and a positive parasitological test (microscopy or RDT) but with no features of severe malaria is defined as having uncomplicated malaria (see section 9.1 for definition of severe falciparum malaria).

Therapeutic objectives
The clinical objectives of treating uncomplicated malaria are to cure the infection as rapidly as possible and to prevent progression to severe disease. “Cure” is defined as elimination of all parasites from the body. The public health objectives of treatment are to prevent onward transmission of the infection to others and to prevent the emergence and spread of resistance to antimalarial drugs.

Incorrect approaches to treatment

Use of monotherapy

The continued use of artemisinins or any of the partner medicines alone will compromise the value of ACT by selecting for drug resistance.

As certain patient groups, such as pregnant women, may need specifically tailored combination regimens, single artemisinin derivatives will still be used in selected referral facilities in the public sector, but they should be withdrawn entirely from the private and informal sectors and from peripheral public health care facilities.

Similarly, continued availability of amodiaquine, mefloquine and SP as monotherapies in many countries is expected to shorten their useful therapeutic life as partner drugs of ACT, and they should be withdrawn wherever possible.

Incomplete dosing

In endemic regions, some semi-immune malaria patients are cured by an incomplete course of antimalarial drugs or by a treatment regimen that would be ineffective in patients with no immunity. In the past, this led to different recommendations for patients considered semi-immune and those considered non-immune. As individual immunity can vary considerably, even in areas of moderate-to-high transmission intensity, this practice is no longer recommended. A full treatment course with a highly effective ACT is required whether or not the patient is considered to be semi-immune.

Another potentially dangerous practice is to give only the first dose of a treatment course to patients with suspected but unconfirmed malaria, with the intention of giving the full treatment if the diagnosis is confirmed. This practice is unsafe, could engender resistance, and is not recommended.

Additional considerations for clinical management

Can the patient take oral medication?

Some patients cannot tolerate oral treatment and will require parenteral or rectal administration for 1–2 days, until they can swallow and retain oral medication reliably. Although such patients do not show other signs of severity, they should receive the same initial antimalarial treatments recommended for severe malaria. Initial rectal or parenteral treatment must always be followed by a full 3-day course of ACT.

Use of antipyretics

In young children, high fevers are often associated with vomiting, regurgitation of medication and seizures. They are thus treated with antipyretics and, if necessary, fanning and tepid sponging. Antipyretics should be used if the core temperature is > 38.5 ºC. Paracetamol (acetaminophen) at a dose of 15 mg/kg bw every 4 h is widely used; it is safe and well tolerated and can be given orally or as a suppository. Ibuprofen (5 mg/kg bw) has been used successfully as an alternative in the treatment of malaria and other childhood fevers, but, like aspirin and other non-steroidal anti-inflammatory drugs, it is *no longer recommended* because of the risks of gastrointestinal bleeding, renal impairment and Reye’s syndrome.

Use of anti-emetics

Vomiting is common in acute malaria and may be severe. Parenteral antimalarial treatment may therefore be required until oral administration is tolerated. Then a full 3-day course of ACT should be given. Anti-emetics are potentially sedative and may have neuropsychiatric adverse effects, which could mask or confound the diagnosis of severe malaria. They should therefore be used with caution.

Management of seizures

Generalized seizures are more common in children with *P. falciparum* malaria than in those with malaria due to other species. This suggests an overlap between the cerebral pathology resulting from falciparum malaria and febrile convulsions. As seizures may be a prodrome of cerebral malaria, patients who have more than two seizures within a 24 h period should be treated as for severe malaria. If the seizures continue, the airways should be maintained and anticonvulsants given (parenteral or rectal benzodiazepines or intramuscular paraldehyde). When the seizure has stopped, the child should be treated as indicated in section 7.10.5, if his or her core temperature is > 38.5 ºC. There is no evidence that prophylactic anticonvulsants are beneficial in otherwise uncomplicated malaria, and they are not recommended.

5.2.1.1 Artemisinin-based combination therapy

Strong recommendation for , High certainty evidence

Artemisinin-based combination therapy (2015)

Children and adults with uncomplicated *P. falciparum* malaria should be treated with one of the following ACTs*:

- artemether-lumefantrine (AL)
- artesunate-amodiaquine (AS+AQ)
- artesunate-mefloquine (ASMQ)
- dihydroartemisinin-piperaquine (DHAP)
- artesunate + sulfadoxine-pyrimethamine (AS+SP)
- artesunate-pyronaridine (ASPY) (2022)

*Artesunate + sulfadoxine-pyrimethamine and artesunate-pyronaridine are not recommended for use in the first trimester of pregnancy. For details of treatment using ACTs in the first trimester of pregnancy, see 5.2.1.4.1 below.

Artesunate-pyronaridine is now included in the list of options for the treatment of uncomplicated malaria (2022). See the full recommendation and supporting evidence below.

Practical info

The pipeline for new antimalarial drugs is healthier than ever before, and several new compounds are in various stages of development. Some novel antimalarial agents are already registered in some countries. The decision to recommend antimalarial drugs for general use depends on the strength of the evidence for safety and efficacy and the context of use. In general, when there are no satisfactory alternatives, newly registered drugs may be recommended; however, for global or unrestricted recommendations, considerably more evidence than that submitted for registration is usually required, to provide sufficient confidence for their safety, efficacy and relative merits as compared with currently recommended treatments.

Several new antimalarial drugs or new combinations have been introduced recently. Some are still in the pre-registration phase and are not discussed here. Arterolane + piperaquine, artemisinin + piperaquine base and artemisinin + napththoquine are new ACTs, which are registered and used in some countries. In addition, there are several new generic formulations of existing drugs. None of these yet has a sufficient evidence base for general recommendation (i.e. unrestricted use).

Arterolane + piperaquine is a combination of a synthetic ozonide and piperaquine phosphate that is registered in India. There are currently insufficient data to make general recommendations.

Artemisinin + piperaquine base combines two well-established, well-tolerated compounds. It differs from previous treatments in that the piperaquine is in the base form, the artemisinin dose is relatively low, and the current recommendation is for only a 2-day regimen. There are insufficient data from clinical trials for a general recommendation, and there is concern that the artemisinin dose regimen provides insufficient protection against resistance to the piperaquine component.

Artemisinin + naphthoquine is also a combination of two relatively old compounds that is currently being promoted as a single-dose regimen, contrary to WHO advice for 3 days of the artemisinin derivative. There are currently insufficient data from rigorously conducted randomized controlled trials to make general recommendations.

Many ACTs are generics. The bioavailability of generics of currently recommended drugs must be comparable to that of the established, originally registered product, and the satisfactory pharmaceutical quality of the product must be maintained.

Please refer to *Good procurement practices for artemisinin-based antimalaria medicines [214].*

Evidence to decision

Benefits and harms	Recommendation: Treat adults and children with uncomplicated *P. falciparum* malaria (including infants, pregnant women in their second and third trimesters and breastfeeding women) with an ACT.

Desirable effects

- Studies have consistently demonstrated that the six WHO-recommended ACTs result in < 5% PCR-adjusted treatment failures in settings with no resistance to the partner drug (high- quality evidence).

Undesirable effects

- Increased cost.

Recommendation: Dihydroartemisinin + piperaquine is recommended for general use.

Desirable effects:

- A PCR-adjusted treatment failure rate of < 5% has been seen consistently in trials of dihydroartemisinin + piperaquine (high-quality evidence).
- Dihydroartemisinin + piperaquine has a longer half-life than artemether + lumefantrine, and fewer new infections occur within 9 weeks of treatment with dihydroartemisinin + piperaquine (high-quality evidence).
- Dihydroartemisinin + piperaquine and artesunate + mefloquine have similar half-lives, and a similar frequency of new infections is seen within 9 weeks of treatment (moderate-quality evidence).

Undesirable effects:

- A few more patients receiving dihydroartemisinin + piperaquine than those given artesunate + mefloquine had a prolonged QT interval (low-quality evidence)
- A few more patients receiving dihydroartemisinin + piperaquine than those given artesunate + mefloquine or artemether + lumefantrine had borderline QT prolongation.

Certainty of the evidence High

For all critical outcomes: High.

Justification

GRADE

In the absence of resistance to the partner drug, the five recommended ACTs have all been shown to achieve a PCR- adjusted treatment failure rate of 5% in many trials in several settings in both adults and children (high-quality evidence) *[215][216]*.

Other considerations

The guideline development group decided to recommend a menu of approved combinations, from which countries can select first- and second-line treatment.

Remarks

Recommendation: Treat adults and children with uncomplicated *P. falciparum* malaria (including infants, pregnant women in their second and third trimesters and breastfeeding women) with ACT.

The WHO-approved first-line ACT options are: artemether + lumefantrine, artesunate + amodiaquine, artesunate + mefloquine, dihydroartemisinin + piperaquine and artesunate + sulfadoxine–pyrimethamine.

These options are recommended for adults and children, including infants, lactating women and pregnant women in their second and third trimester.

In deciding which ACTs to adopt in national treatment policies, national policy- makers should take into account: the pattern of resistance to antimalarial drugs in the country, the relative efficacy and safety of the combinations, their cost, the availability of paediatric formulations and the availability of co-formulated products.

Fixed-dose combinations are preferred to loose tablets or co-blistered products.

The Guideline Development Group decided to recommend a “menu” of approved combinations from which countries can select first- and second- line therapies. Modelling studies suggest that having multiple first-line ACTs available for use may help to prevent or delay the development of resistance.

Recommendation: Dihydroartemisinin + piperaquine is recommended for general use.

A systematic review showed that the dosing regimen of dihydroartemisinin + piperaquine currently recommended by the manufacturers leads to sub-optimal dosing in young children. The group plans to recommend a revised dosing regimen based on models of pharmacokinetics.

Further studies of the risk for QT interval prolongation have been requested by the European Medicines Agency.

ACT is a combination of a rapidly acting artemisinin derivative with a longer-acting (more slowly eliminated) partner drug. The artemisinin component rapidly clears parasites from the blood (reducing parasite numbers by a factor of approximately 10 000 in each 48 h asexual cycle) and is also active against the sexual stages of the gametocytes that mediate onward transmission to mosquitos. The longer- acting partner drug clears the remaining parasites and provides protection against development of resistance to the artemisinin derivative. Partner drugs with longer elimination half-lives also provide a period of post-treatment prophylaxis.

The GDG recommended dihydroartemisinin + piperaquine for use in 2009 but re-evaluated the evidence in 2013 because additional data on its safety had become available. The group noted the small absolute prolongation of the QT interval with dihydroartemisinin + piperaquine but was satisfied that the increase was of comparable magnitude to that observed with chloroquine and was not important clinically *[214][217]*.

Strong recommendation for , Low certainty evidence

Artesunate-pyronaridine for uncomplicated malaria (2022)

Artesunate-pyronaridine (ASPY) is recommended as an artemisinin-based combination therapy option for the treatment of uncomplicated *P. falciparum* malaria.

- *ASPY should be avoided by individuals with known liver disease (clinically apparent liver disease) because ASPY is associated with liver transaminitis.*
- *Pharmacovigilance should be strengthened where ASPY is used for the treatment of malaria.*

Practical info

As with the deployment of any new malaria treatment, pharmacovigilance and resistance surveillance systems should be strengthened.

Evidence to decision

Benefits and harms	• ASPY, with large treatment effects, has been shown to be non-inferior in efficacy compared to the currently recommended ACTs. The overall benefit of this additional ACT is its potential to provide an alternative treatment, thereby reducing pressure on the partner medicines in the face of emerging artemisinin partner drug resistance. • Compared to other ACTs, ASPY may have fewer PCR-adjusted and PCR-unadjusted failures at day 28, while results for day 42 are inconclusive. Data for children are limited. • Following careful safety reviews, the conclusions are that the use of ASPY can be accompanied by mild, reversible and asymptomatic elevations of some liver enzymes, but that these elevations are not associated with clinically detected hepatotoxicity. ASPY is more likely than artemether-lumefantrine or artesunate-mefloquine to increase aspartate transaminase (AST) and alanine aminotransferase (ALT) > 5 times, but the risks are similar to those of artesunate-amodiaquine; there is no clear association of ASPY with increased bilirubin. There is no evidence to date to suggest that these transiently elevated transaminases result in serious liver injury. • The risk of vomiting appears to be significantly higher in young children (7.7%) and infants (11.2%) than in older children (3.1%) and adults (2.8%) *[218]*. However, the overall risk of vomiting with ASPY is similar to the risks with other ACTs (OR: 0.91; 95% CI: 0.71–1.17; nine studies; n=5534) *[219]*.

- There are no data available from patients with pre-existing liver conditions (e.g. hepatitis B or C) or from those with risk factors for liver disease (e.g. receiving medicines known to be hepatotoxic, use of potentially hepatotoxic herbal medicines, alcohol abuse). There is, however, some early reassuring data from a study *[218]* that included limited data on inadvertent exposures of patients with HIV (15 exposures) and 158 persons with elevated liver enzymes (AST or ALT > 2 times the upper limit of normal [ULN]) at baseline. Caution is advised in these patients when considering ASPY as treatment, as these risk factors, as well as coadministration of potential hepatotoxic medicines (including paracetamol commonly used in patients with malaria), might have a cumulative adverse effect on the liver.

Certainty of the evidence

Low

The GDG judged the overall certainty of the assessed evidence to be low mostly due to imprecision and indirectness.

Values and preferences

The GDG determined that there was probably no important uncertainty or variability in individual patients' values and preferences, but country-level value judgements are still important, as these could be influenced by the prevalence of antimalarial partner drug resistance and the prevalence of hepatic diseases.

Resources

Research evidence
Research on formal cost analysis, and cost estimates related to scale are required. However, changing first- or second-line malaria treatment is quite resource-intensive, requiring staff training and patient information and introducing supply chain and logistical issues. However, introducing ASPY is not expected to be different from other ACTs already in use, as any additional cost would be minimal based on the actual cost of the medicine.

Summary
Research on formal cost analysis, and cost estimates related to scale are required. However, changing first- or second-line malaria treatment is quite resource-intensive, requiring staff training and patient information and introducing supply chain and logistical issues. However, introducing ASPY is not expected to be different from other ACTs already in use, as any additional cost would be minimal based on the actual cost of the medicine.

Equity

The GDG considered that ASPY is likely to enhance equity, especially in areas of emerging resistance to existing combinations. The addition of ASPY as a treatment option for malaria will probably increase health equity.

Acceptability

Although in some countries there is limited experience of its use, ASPY is probably acceptable given that some countries already include ASPY in therapeutic efficacy studies. Aside from the additional resource implications with the introduction of a new antimalarial regimen, oral treatments are generally well accepted. Some issues might arise when hepatic risk profiles need to be assessed in the target population.

Feasibility

Policy changes are feasible, since some countries have already started to use ASPY. The medicine is available, and the treatment regimen is similar to that of other approved ACTs. ASPY has also received a positive scientific opinion from the European Medicines Agency under Article 58 and is thus included in the WHO list of prequalified antimalarial medicines. However, the feasibility of strengthening pharmacovigilance will be highly variable from country to country.

Justification

The GDG reached a consensus on a strong recommendation for the intervention, despite the low certainty of evidence because of:

- the large magnitude of treatment effect, as well as its non-inferiority and comparability to the other currently recommended ACTs;
- its tolerability and generally mild, reversible adverse events; and
- the probable increased equity from access to an additional treatment option, specifically in the face of increasing ACT partner drug resistance.

Research needs

The GDG highlighted the following evidence gaps requiring further research. These relate to:

- individual patient data meta-analysis comparing hepatic safety and gastrointestinal tolerability (particularly vomiting in young children and infants within one hour of dosing, as this could alter efficacy) between ASPY and other ACTs;
- continued assessment of efficacy, safety and tolerability of all ACTs, including ASPY, across malaria-endemic regions, especially in African children;
- further monitoring of efficacy, particularly in children in different settings, and monitoring for adverse events from inadvertent pregnancy exposures; and
- identification and validation of molecular markers of resistance to pyronaridine.

5.2.1.1.1 Duration of treatment

A 3-day course of the artemisinin component of ACTs covers two asexual cycles, ensuring that only a small fraction of parasites remain for clearance by the partner drug, thus reducing the potential development of resistance to the partner drug. Shorter courses (1–2 days) are therefore not recommended, as they are less effective, have less effect on gametocytes and provide less protection for the slowly eliminated partner drug.

Strong recommendation for , High certainty evidence

Duration of ACT treatment (2015)

ACT regimens should provide 3 days' treatment with an artemisinin derivative.

Evidence to decision

Benefits and harms	**Desirable effects** • Fewer patients taking ACTs containing 3 days of an artemisinin derivative experience treatment failure within the first 28 days (high-quality evidence). • Fewer participants taking ACTs containing 3 days of an artemisinin derivative have gametocytaemia at day 7 (high-quality evidence).
Certainty of the evidence	High For all critical outcomes: High.

Justification

GRADE

In four randomized controlled trials in which the addition of 3 days of artesunate to SP was compared directly with 1 day of artesunate with SP:

Three days of artesunate reduced the PCR-adjusted treatment failure rate within the first 28 days from that with 1 day of artesunate (RR, 0.45; 95% CI, 0.36–0.55, four trials, 1202 participants, high-quality evidence).

Three days of artesunate reduced the number of participants who had gametocytaemia at day 7 from that with 1 day of artesunate (RR, 0.74; 95% CI, 0.58–0.93, four trials, 1260 participants, high-quality evidence).

Other considerations
The guideline development group considered that 3 days of artemisinin derivative are necessary to provide sufficient efficacy, promote good adherence and minimize the risk of drug resistance resulting from incomplete treatment.

Remarks
Longer ACT treatment may be required to achieve > 90% cure rate in areas with artemisinin-resistant *P. falciparum*, but there are insufficient trials to make definitive recommendations. A 3-day course of the artemisinin component of ACTs covers two asexual cycles, ensuring that only a small fraction of parasites remain for clearance by the partner drug, thus reducing the potential development of resistance to the partner drug. Shorter courses (1–2 days) are therefore not recommended, as they are less effective, have less effect on gametocytes and provide less protection for the slowly eliminated partner drug.

Rationale for the recommendation
The Guideline Development Group considers that 3 days of an artemisinin derivative are necessary to provide sufficient efficacy, promote good adherence and minimize the risk for drug resistance due to incomplete treatment.

5.2.1.1.2 Dosing of ACTs

ACT regimens must ensure optimal dosing to prolong their useful therapeutic life, i.e. to maximize the likelihood of rapid clinical and parasitological cure, minimize transmission and retard drug resistance.

It is essential to achieve effective antimalarial drug concentrations for a sufficient time (exposure) in all target populations in order to ensure high cure rates. The dosage recommendations below are derived from understanding the relationship between dose and the profiles of exposure to the drug (pharmacokinetics) and the resulting therapeutic efficacy (pharmacodynamics) and safety. Some patient groups, notably younger children, are not dosed optimally with the "dosage regimens recommended by manufacturers, which compromises efficacy and fuels resistance. In these guidelines when there was pharmacological evidence that certain patient groups are not receiving optimal doses, dose regimens were adjusted to ensure similar exposure across all patient groups.

Weight-based dosage recommendations are summarized below. While age-based dosing may be more practical in children, the relation between age and weight differs in different populations. Age-based dosing can therefore result in under- dosing or over-dosing of some patients, unless large, region-specific weight-for-age databases are available to guide dosing in that region.

Factors other than dosage regimen may also affect exposure to a drug and thus treatment efficacy. The drug exposure of an individual patient also depends on factors such as the quality of the drug, the formulation, adherence and, for some drugs, co-administration with fat. Poor adherence is a major cause of treatment failure and drives the emergence and spread of drug resistance. Fixed-dose combinations encourage adherence and are preferred to loose (individual) tablets. Prescribers should take the time necessary to explain to patients why they should complete antimalarial course.

Artemether + lumefantrine
Formulations currently available: Dispersible or standard tablets containing 20 mg artemether and 120 mg lumefantrine, and standard tablets containing 40 mg artemether and 240 mg lumefantrine in a fixed-dose combination formulation. The flavoured dispersible tablet paediatric formulation facilitates use in young children.

Target dose range: A total dose of 5–24 mg/kg bw of artemether and 29–144 mg/ kg bw of lumefantrine

Recommended dosage regimen: Artemether + lumefantrine is given twice a day for 3 days (total, six doses). The first two doses should, ideally, be given 8 h apart.

Body weight (kg)	Dose (mg) of artemether + lumefantrine given twice daily for 3 days
< 15	20 + 120
15 to < 25	40 + 240
25 to < 35	60 + 360

≥ 35	80 + 480

Factors associated with altered drug exposure and treatment response:

- Decreased exposure to lumefantrine has been documented in young children (<3 years) as well as pregnant women, large adults, patients taking mefloquine, rifampicin or efavirenz and in smokers. As these target populations may be at increased risk for treatment failure, their responses to treatment should be monitored more closely and their full adherence ensured.
- Increased exposure to lumefantrine has been observed in patients concomitantly taking lopinavir- lopinavir/ritonavir-based antiretroviral agents but with no increase in toxicity; therefore, no dosage adjustment is indicated.

Additional comments:

- An advantage of this ACT is that lumefantrine is not available as a monotherapy and has never been used alone for the treatment of malaria.
- Absorption of lumefantrine is enhanced by co-administration with fat. Patients or caregivers should be informed that this ACT should be taken immediately after food or a fat containing drink (e.g. milk), particularly on the second and third days of treatment.

Artesunate + amodiaquine
Formulations currently available: A fixed-dose combination in tablets containing 25 + 67.5 mg, 50 + 135 mg or 100 + 270 mg of artesunate and amodiaquine, respectively

Target dose and range: The target dose (and range) are 4 (2–10) mg/kg bw per day artesunate and 10 (7.5–15) mg/kg bw per day amodiaquine once a day for 3 days. A total therapeutic dose range of 6–30 mg/kg bw per day artesunate and 22.5–45 mg/kg bw per dose amodiaquine is recommended.

Body weight (kg)	Artesunate + amodiaquine dose (mg) given daily for 3 days
< 9	25 + 67.5
9 to < 18	50 + 135
18 to < 36	100 + 270
≥ 36	200 + 540

Factors associated with altered drug exposure and treatment response:

Treatment failure after amodiaquine monotherapy was more frequent among children who were underweight for their age. Therefore, their response to artesunate + amodiaquine treatment should be closely monitored.

Artesunate + amodiaquine is associated with severe neutropenia, particularly in patients co-infected with HIV and especially in those on zidovudine and/or cotrimoxazole. Concomitant use of efavirenz increases exposure to amodiaquine and hepatotoxicity. Thus, concomitant use of artesunate + amodiaquine by patients taking zidovudine, efavirenz and cotrimoxazole should be avoided, unless this is the only ACT promptly available.

Additional comments:

No significant changes in the pharmacokinetics of amodiaquine or its metabolite desethylamodiaquine have been observed during the second and third trimesters of pregnancy; therefore, no dosage adjustments are recommended.

No effect of age has been observed on the plasma concentrations of amodiaquine and desethylamodiaquine, so no dose adjustment by age is indicated. Few data are available on the pharmacokinetics of amodiaquine in the first year of life.

Artesunate + mefloquine
Formulations currently available: A fixed-dose formulation of paediatric tablets containing 25 mg artesunate and 55 mg mefloquine hydrochloride (equivalent to 50 mg mefloquine base) and adult tablets containing 100 mg artesunate and 220 mg mefloquine hydrochloride (equivalent to 200 mg mefloquine base)

Target dose and range: Target doses (ranges) of 4 (2–10) mg/kg bw per day artesunate and 8.3 (7–11) mg/kg bw per day mefloquine, given once a day for 3 days

Body weight (kg)	Artesunate + mefloquine dose (mg) given daily for 3 days
< 9	25 + 55
9 to < 18	50 + 110
18 to < 30	100 + 220
≥ 30	200 + 440

Additional comments:

Mefloquine was associated with increased incidences of nausea, vomiting, dizziness, dysphoria and sleep disturbance in clinical trials, but these symptoms are seldom debilitating, and, where this ACT has been used, it has generally been well tolerated. To reduce acute vomiting and optimize absorption, the total mefloquine dose should preferably be split over 3 days, as in current fixed-dose combinations.

As concomitant use of rifampicin decreases exposure to mefloquine, potentially decreasing its efficacy, patients taking this drug should be followed up carefully to identify treatment failures.

Artesunate + sulfadoxine–pyrimethamine
Formulations: Currently available as blister-packed, scored tablets containing 50 mg artesunate and fixed dose combination tablets comprising 500 mg sulfadoxine + 25 mg pyrimethamine. There is no fixed-dose combination.

Target dose and range: A target dose (range) of 4 (2–10) mg/kg bw per day artesunate given once a day for 3 days and a single administration of at least 25 / 1.25 (25–70 / 1.25–3.5) mg/kg bw sulfadoxine / pyrimethamine given as a single dose on day 1.

Body weight (kg)	Artesunate dose given daily for 3 days (mg)	Sulfadoxine / pyrimethamine dose (mg) given as a single dose on day 1
< 10	25 mg	250 / 12.5
10 to < 25	50 mg	500 / 25
25 to < 50	100 mg	1000 / 50
≥ 50	200 mg	1500 / 75

Factors associated with altered drug exposure and treatment response: The low dose of folic acid (0.4 mg daily) that is required to protect the fetuses of pregnant women from neural tube defects do not reduce the efficacy of SP, whereas higher doses (5 mg daily) do significantly reduce its efficacy and should not be given concomitantly.

Additional comments:

The disadvantage of this ACT is that it is not available as a fixed-dose combination. This may compromise adherence and increase the risk for distribution of loose artesunate tablets, despite the WHO ban on artesunate monotherapy.

Resistance is likely to increase with continued widespread use of SP, sulfalene– pyrimethamine and cotrimoxazole (trimethoprim-sulfamethoxazole). Fortunately, molecular markers of resistance to antifols and sulfonamides correlate well with therapeutic responses. These should be monitored in areas in which this drug is used.

Strong recommendation for

Revised dose recommendation for dihydroartemisinin + piperaquine in young children (2015)

Children weighing <25kg treated with dihydroartemisinin + piperaquine should receive a minimum of 2.5 mg/kg bw per day of dihydroartemisinin and 20 mg/ kg bw per day of piperaquine daily for 3 days.

*Not evaluated using the GRADE framework

Practical info

Formulations: Currently available as a fixed-dose combination in tablets containing 40 mg dihydroartemisinin and 320 mg piperaquine and paediatric tablets contain 20 mg dihydroartemisinin and 160 mg piperaquine.

Target dose and range: A target dose (range) of 4 (2–10) mg/kg bw per day dihydroartemisinin and 18 (16–27) mg/kg bw per day piperaquine given once a day for 3 days for adults and children weighing ≥ 25 kg. The target doses and ranges for children weighing < 25 kg are 4 (2.5–10) mg/kg bw per day dihydroartemisinin and 24 (20–32) mg/kg bw per day piperaquine once a day for 3 days.

Recommended dosage regimen: The dose regimen currently recommended by the manufacturer provides adequate exposure to piperaquine and excellent cure rates (> 95%), except in children < 5 years, who have a threefold increased risk for treatment failure. Children in this age group have significantly lower plasma piperaquine concentrations than older children and adults given the same mg/kg bw dose. Children weighing < 25 kg should receive at least 2.5 mg/kg bw dihydroartemisinin and 20 mg/kg bw piperaquine to achieve the same exposure as children weighing ≥ 25 kg and adults.

Dihydroartemisinin + piperaquine should be given daily for 3 days.

Body weight (kg)	Dihydroartemisinin + piperaquine dose (mg) given daily for 3 days
< 8	20 + 160
8 to < 11	30 + 240
11 to < 17	40 + 320
17 to < 25	60 + 480
25 to < 36	80 + 640
36 to < 60	120 + 960
60 < 80	160 + 1280
>80	200 + 1600

Factors associated with altered drug exposure and treatment response:

High-fat meals should be avoided, as they significantly accelerate the absorption of piperaquine, thereby increasing the risk for potentially arrhythmogenic delayed ventricular repolarization (prolongation of the corrected electrocardiogram QT interval). Normal meals do not alter the absorption of piperaquine.

As malnourished children are at increased risk for treatment failure, their response to treatment should be monitored closely.

- Dihydroartemisinin exposure is lower in pregnant women.
- Piperaquine is eliminated more rapidly by pregnant women, shortening the post-treatment prophylactic effect of dihydroartemisinin + piperaquine. As this does not affect primary efficacy, no dosage adjustment is recommended for pregnant women.

Additional comments: Piperaquine prolongs the QT interval by approximately the same amount as chloroquine but by less than quinine. It is not necessary to perform an electrocardiogram before prescribing dihydroartemisinin + piperaquine, but this ACT should not be used in patients with congenital QT prolongation or who have a clinical condition or are on medications that prolong the QT interval. There has been no evidence of cardiotoxicity in large randomized trials or in extensive deployment.

Justification

The dosing subgroup reviewed all available dihydroartemisinin-piperaquine pharmacokinetic data (6 published studies and 10 studies from the WWARN database; total 652 patients) *[208][211]* and then conducted simulations of piperaquine exposures for each weight group. These showed lower exposure in younger children with higher risks of treatment failure. The revised dose regimens are predicted to provide equivalent piperaquine exposures across all age groups.

Other considerations
This dose adjustment is not predicted to result in higher peak piperaquine concentrations than in older children and adults, and as there is no evidence of increased toxicity in young children, the GRC concluded that the predicted benefits of improved antimalarial exposure are not at the expense of increased risk.

5.2.1.2 Recurrent falciparum malaria

Recurrence of *P. falciparum* malaria can result from re-infection or recrudescence (treatment failure). Treatment failure may result from drug resistance or inadequate exposure to the drug due to sub-optimal dosing, poor adherence, vomiting, unusual pharmacokinetics in an individual, or substandard medicines. It is important to determine from the patient's history whether he or she vomited the previous treatment or did not complete a full course of treatment.

When possible, treatment failure must be confirmed parasitologically. This may require referring the patient to a facility with microscopy or LDH-based RDTs, as *P. falciparum* histidine-rich protein-2 (*Pf*HRP2)-based tests may remain positive for weeks after the initial infection, even without recrudescence. Referral may be necessary anyway to obtain second-line treatment. In individual patients, it may not be possible to distinguish recrudescence from re-infection, although lack of resolution of fever and parasitaemia or their recurrence within 4 weeks of treatment are considered failures of treatment with currently recommended ACTs. In many cases, treatment failures are missed because patients are not asked whether they received antimalarial treatment within the preceding 1–2 months. Patients who present with malaria should be asked this question routinely.

Failure within 28 days
The recommended second-line treatment is an alternative ACT known to be effective in the region. Adherence to 7-day treatment regimens (with artesunate or quinine both of which should be co-administered with + tetracycline, or doxycycline or clindamycin) is likely to be poor if treatment is not directly observed; these regimens are no longer generally recommended. The distribution and use of oral artesunate monotherapy outside special centres are strongly discouraged, and quinine-containing regimens are not well tolerated.

Failure after 28 days
Recurrence of fever and parasitaemia > 4 weeks after treatment may be due to either recrudescence or a new infection. The distinction can be made only by PCR genotyping of parasites from the initial and the recurrent infections.

As PCR is not routinely used in patient management, all presumed treatment failures after 4 weeks of initial treatment should, from an operational standpoint, be considered new infections and be treated with the first-line ACT. However, reuse of mefloquine within 60 days of first treatment is associated with an increased risk for neuropsychiatric reactions, and an alternative ACT should be used.

5.2.1.3 Reducing the transmissibility of treated P. falciparum infections in areas of low-intensity transmission

Strong recommendation for , Low certainty evidence

Reducing the transmissibility of treated *P. falciparum* infections (2024)

In low-transmission areas, a single dose of 0.25 mg/kg bw primaquine should be given with an ACT to patients with *P. falciparum* malaria (except pregnant women) to reduce transmission. G6PD testing is not required.

Practical info

In light of concern about the safety of the previously recommended dose of 0.75 mg/kg bw in individuals with G6PD deficiency, a WHO panel reviewed the safety of primaquine as a *P. falciparum* gametocytocide and concluded that a single dose of 0.25 mg/kg bw of primaquine base is unlikely to cause serious toxicity, even in people with G6PD deficiency *[223]*. Thus, where indicated a single dose of 0.25mg/kg bw of primaquine base should be given on the first day of treatment, in addition to an ACT, to all patients with parasitologically confirmed *P. falciparum* malaria except for pregnant women, infants < 1 months of age and women breastfeeding infants < 1 months of age. Secretion of primaquine in breast milk is negligible *[224]*.

Dosing table based on the most widely currently available tablet strength (7.5mg base)

Body weight (kg)	Single dose of primaquine (mg base)
5 to < 25[a]	3.75
25 to < 50	7.5
50 to 100	15

[a] Dosing of young children weighing < 10 kg is limited by the tablet sizes currently available.

Please refer to the *Policy brief on single-dose primaquine as a gametocytocide in Plasmodium falciparum malaria* *[225]*.

Evidence to decision

Benefits and harms

Desirable effects

- Single doses of primaquine > 0.4 mg/kg bw reduced gametocyte carriage at day 8 by around two thirds (moderate-quality evidence).
- There are too few trials of doses < 0.4 mg/kg bw to quantify the effect on gametocyte carriage (low-quality evidence).
- Analysis of observational data from mosquito feeding studies suggests that 0.25 mg/kg bw may rapidly reduce the infectivity of gametocytes to mosquitoes.

Undesirable effects

- People with severe G6PD deficiency are at risk for haemolysis. At this dose, however, the risk is thought to be small; there are insufficient data to quantify this risk.

Certainty of the evidence

Low

Overall certainty of evidence for all critical outcomes: low.

Justification

GRADE

In an analysis of observational studies of single-dose primaquine, data from mosquito feeding studies on 180 people suggest that adding 0.25 mg/kg primaquine to treatment with an ACT can rapidly reduce the infectivity of gametocytes to mosquitoes.

In a systematic review of eight randomized controlled trials of the efficacy of adding single-dose primaquine to ACTs for reducing the transmission of malaria, in comparison with ACTs alone *[226]*:

- single doses of > 0.4 mg/kg bw primaquine reduced gametocyte carriage at day 8 by about two thirds (RR, 0.34; 95% CI, 0.19–0.59, two trials, 269 participants, *high-certainty evidence*); and
- single doses of primaquine > 0.6 mg/kg bw reduced gametocyte carriage at day 8 by about two thirds (RR, 0.29; 95% CI, 0.22–0.37, seven trials, 1380 participants, *high-certainty evidence*).

There have been no randomized controlled trials of the effects on the incidence of malaria or on transmission to mosquitos.

Other considerations
The guideline development group considered that the evidence of a dose– response relation from observational studies of mosquito feeding was sufficient to conclude the primaquine dose of 0.25mg/kg bw significantly reduced *P. falciparum* transmissibility.

The population benefits of reducing malaria transmission with gametocytocidal drugs such as primaquine require that a very high proportion of treated patients receive these medicines and that there is no large transmission reservoir of asymptomatic parasite carriers. This strategy is therefore likely to be effective only in areas of low-intensity malaria transmission, as a component of elimination programmes.

Remarks
This recommendation excludes high-transmission settings, as symptomatic patients make up only a small proportion of the total population carrying gametocytes within a community, and primaquine is unlikely to affect transmission.

A major concern of national policy-makers in using primaquine has been the small risk for haemolytic toxicity in G6PD-deficient people, especially where G6PD testing is not available.

Life-threatening haemolysis is considered unlikely with the 0.25mg/kg bw dose and without G6PD testing *[227]*.

Rationale for the recommendation
The Guideline Development Group considered the evidence on dose–response relations in the observational mosquito-feeding studies of reduced transmissibility with the dose of 0.25 mg/kg bw and the judgement of the WHO Evidence Review Group (November 2012). Their view was that the potential public health benefits of single low-dose (0.25 mg/kg bw) primaquine in addition to an ACT for falciparum malaria, without G6PD testing, outweigh the potential risk for adverse effects.

5.2.1.4 Special risk groups

Several important patient sub-populations, including young children, pregnant women and patients taking potent enzyme inducers (e.g. rifampicin, efavirenz), have altered pharmacokinetics, resulting in sub-optimal exposure to antimalarial drugs. This increases the rate of treatment failure with current dosage regimens. The rates of treatment failure are substantially higher in hyperparasitaemic patients and patients in areas with artemisinin-resistant falciparum malaria, and these groups require greater exposure to antimalarial drugs (longer duration of therapeutic concentrations) than is achieved with current ACT dosage recommendations. It is often uncertain how best to achieve this. Options include increasing individual doses, changing the frequency or duration of dosing, or adding an additional antimalarial drug. Increasing individual doses may not, however, achieve the desired exposure (e.g., lumefantrine absorption becomes saturated), or the dose may be toxic due to transiently high plasma concentrations (piperaquine, mefloquine, amodiaquine, pyronaridine). An additional advantage of lengthening the duration of treatment (by giving a 5-day regimen) is that it provides additional exposure of the asexual cycle to the artemisinin component as well as augmenting exposure to the partner drug. The acceptability, tolerability, safety and effectiveness of augmented ACT regimens in these special circumstances should be evaluated urgently.

Large and obese adults
Large adults are at risk for under-dosing when they are dosed by age or in standard pre-packaged adult weight-based treatments. In principle, dosing of large adults should be based on achieving the target mg/kg bw dose for each antimalarial regimen. The practical consequence is that two packs of an antimalarial drug might have to be opened to ensure adequate treatment. For obese patients, less drug is often distributed to fat than to other tissues; therefore, they should be dosed on the basis of an estimate of lean body weight, ideal body weight. Patients who are heavy but not obese require the same mg/kg bw doses as lighter patients.

In the past, maximum doses have been recommended, but there is no evidence or justification for this practice. As the evidence for an association between dose, pharmacokinetics and treatment outcome in overweight or large adults is limited, and alternative dosing options have not been assessed in treatment trials, it is recommended that this gap in knowledge be assessed urgently. In the absence of data, treatment providers should attempt to follow up the treatment outcomes of large adults whenever possible.

5.2.1.4.1 Pregnant and lactating women

Malaria in pregnancy is associated with low-birth-weight infants, increased anaemia and, in low-transmission areas, increased risks for severe malaria, pregnancy loss and death. In high-transmission settings, despite the adverse effects on fetal growth, malaria is usually asymptomatic in pregnancy or is associated with only mild, non-specific symptoms. There is insufficient information on the safety, efficacy and pharmacokinetics of most antimalarial agents in pregnancy, particularly during the first trimester.

First trimester of pregnancy
Malaria in pregnancy is associated with low birthweight in infants, increased anaemia and, in low-transmission areas, increased risks for severe malaria, pregnancy loss and death. Malaria in pregnancy is, therefore, considered a priority problem. The risk of malaria infection is said to be highest in the first and second trimesters of pregnancy *[226]*. In a study in Benin, the prevalence of malaria infection in the first trimester was 21.8% and was significantly associated with maternal anaemia in the third trimester (adjusted odds ratio [aOR]: 2.25; 95% CI: 1.11–4.55) *[227]*. A modelling study among women in areas of stable malaria transmission suggested that over 60% of malaria infections during pregnancy occur by the end of the first trimester *[228]*.

Although ACTs have been shown to be more effective and better tolerated and provide longer post-treatment prophylaxis than oral quinine in the second and third trimesters of pregnancy, to date, WHO had recommended quinine + clindamycin instead of ACTs for the first trimester. This recommendation was due to concerns about the potential teratogenicity of the artemisinin observed in pre-clinical animal studies *[229][230]*.

WHO has generated a new recommendation based on a review of all updated evidence to date on the risks and benefits of using any ACT compared to quinine for the treatment of uncomplicated *P. falciparum* malaria in the first trimester of pregnancy. The new recommendation is given in the box below.

Second and third trimesters
Experience with artemisinin derivatives in the second and third trimesters (over 4000 documented pregnancies) is increasingly reassuring: no adverse effects on the mother or fetus have been reported. The current assessment of risk–benefit suggests that ACTs should be used to treat uncomplicated falciparum malaria in the second and third trimesters of pregnancy. The current standard six-dose artemether + lumefantrine regimen for the treatment of uncomplicated falciparum malaria has been evaluated in > 1000 women in the second and third trimesters in controlled

trials and has been found to be well tolerated and safe. In a low-transmission setting on the Myanmar–Thailand border, however, the efficacy of the standard six-dose artemether + lumefantrine regimen was inferior to 7 days of artesunate monotherapy. The lower efficacy may have been due to lower drug concentrations in pregnancy, as was also recently observed in a high-transmission area in Uganda and the United Republic of Tanzania. Although many women in the second and third trimesters of pregnancy in Africa have been exposed to artemether + lumefantrine, further studies are under way to evaluate its efficacy, pharmacokinetics and safety in pregnant women. Similarly, many pregnant women in Africa have been treated with amodiaquine alone or combined with SP or artesunate; however, amodiaquine use for the treatment of malaria in pregnancy has been formally documented in only > 1300 pregnancies. Use of amodiaquine in women in Ghana in the second and third trimesters of pregnancy was associated with frequent minor side- effects but not with liver toxicity, bone marrow depression or adverse neonatal outcomes.

Dihydroartemisinin + piperaquine was used successfully in the second and third trimesters of pregnancy in > 2000 women on the Myanmar–Thailand border for rescue therapy and in Indonesia for first-line treatment. SP, although considered safe, is not appropriate for use as an artesunate partner drug in many areas because of resistance to SP. If artesunate + SP is used for treatment, co-administration of daily high doses (5 mg) of folate supplementation should be avoided, as this compromises the efficacy of SP. A lower dose of folate (0.4–0.5 mg bw/day) or a treatment other than artesunate + SP should be used.

Mefloquine is considered safe for the treatment of malaria during the second and third trimesters; however, it should be given only in combination with an artemisinin derivative.

Quinine is associated with an increased risk for hypoglycaemia in late pregnancy, and it should be used (with clindamycin) only if effective alternatives are not available.

Primaquine and tetracyclines should not be used in pregnancy.

Dosing in pregnancy
Data on the pharmacokinetics of antimalarial agents used during pregnancy are limited. Those available indicate that pharmacokinetic properties are often altered during pregnancy but that the alterations are insufficient to warrant dose modifications at this time. With quinine, no significant differences in exposure have been seen during pregnancy. Studies of the pharmacokinetics of SP used in IPTp in many sites show significantly decreased exposure to sulfadoxine, but the findings on exposure to pyrimethamine are inconsistent. Therefore, no dose modification is warranted at this time.

Studies are available of the pharmacokinetics of artemether + lumefantrine, artesunate + mefloquine and dihydroartemisinin + piperaquine. Most data exist for artemether + lumefantrine; these suggest decreased overall exposure during the second and third trimesters. Simulations suggest that a standard six-dose regimen of lumefantrine given over 5 days, rather than 3 days, improves exposure, but the data are insufficient to recommend this alternative regimen at present. Limited data on pregnant women treated with dihydroartemesinin + piperaquine suggest lower dihydroartemisinin exposure and no overall difference in total piperaquine exposure, but a shortened piperaquine elimination half-life was noted. The data on artesunate + mefloquine are insufficient to recommend an adjustment of dosage. No data are available on the pharmacokinetics of artesunate + amodiaquine in pregnant women with falciparum malaria, although drug exposure was similar in pregnant and non-pregnant women with vivax malaria.

Lactating women
The amounts of antimalarial drugs that enter breast milk and are consumed by breastfeeding infants are relatively small. Tetracycline is contraindicated in breastfeeding mothers because of its potential effect on infants' bones and teeth. Pending further information on excretion in breast milk, primaquine should not be used for nursing women, unless the breastfed infant has been checked for G6PD deficiency.

Strong recommendation for , Low certainty evidence

Treatment in the first trimester of pregnancy (2022)

Pregnant women with uncomplicated *P. falciparum* malaria should be treated with artemether-lumefantrine during the first trimester.

- *Limited exposures to other ACTs (artesunate-amodiaquine, artesunate-mefloquine and dihydroartemisinin-piperaquine) suggest that the current evidence is insufficient to make a recommendation for routine use of these other ACTs in the first trimester of pregnancy. However, consistent with the previous WHO recommendation that provided for limited use of ACTs if the first-line recommended medicine was not available, these other ACTs may be considered for use where artemether-lumefantrine is not a recommended ACT for uncomplicated malaria or is not available, given the demonstrated poorer outcomes of quinine treatment, along with the challenges of adherence to a seven-day course of treatment.*
- *Antifolates are contraindicated in the first trimester of pregnancy. Therefore, ACTs containing sulfadoxine-pyrimethamine are contraindicated during the first trimester of pregnancy.*
- *There is currently no documented record of the use of artesunate-pyronaridine during the first trimester of pregnancy.*
- *Continued pharmacovigilance and clinical research, including prospective controlled trials on the efficacy and safety of antimalarial medicines for the treatment of malaria in pregnancy, should be supported and funded.*

Practical info

As with the deployment of any new malaria treatment recommendations, pharmacovigilance and adverse events and pregnancy outcome surveillance systems should be strengthened.

Evidence to decision

Benefits and harms

ACTs have large positive effects with respect to efficacy, effectiveness and tolerability compared to quinine in non-pregnant patients and women in the second and third trimesters of pregnancy. Systematic reviews have shown that treatment failures are six times more likely with quinine than with artemether-lumefantrine in the second and third trimesters of pregnancy *[231][232]*.

In various animal studies (including rodents and monkeys), artemisinin has been found to deplete embryonic erythroblasts at relatively low doses of 1/200–1/400 of the LD50 (equivalent to > 10 mg/kg body weight), leading to malformation or embryonic death *[229][233]*. The adverse effects include embryo resorption, pregnancy loss and congenital anomalies, including shortening of the long bones and heart defects (ventricular septal and great vessel defects) *[234][235]*. For this reason, despite its demonstrated lower efficacy in the second and third trimesters of pregnancy, quinine (in combination with clindamycin) was retained by WHO in 2015 for treatment in the first trimester until adequate numbers of human exposures to artemisinin could allow for more safety assessments in humans.

In weighing the risk–benefit ratio, safety risks from antimalarial treatment need to be weighed against the adverse effects of malaria in the first trimester *[236]*.

A recently updated individual patient data meta-analysis of 34 178 pregnancies included 737 well documented pregnancies exposed to artemisinin and 1076 exposed to non-artemisinin-based treatments in the first trimester. Of the exposures to artemisinin, 71% (525) were to artemether-lumefantrine *[237]*. This meta-analysis provided the basis for the re-evaluation of the treatment of malaria in the first trimester of pregnancy.

This updated individual patient data review showed that first-trimester treatment with artemether-lumefantrine was associated with significantly fewer adverse pregnancy outcomes than first-trimester treatment with quinine. Treatment with artemether-lumefantrine in the first trimester was associated with a statistically significant lower risk (42%) of adverse pregnancy outcomes compared to treatment with oral quinine (aHR: 0.58; 95% CI: 0.36–0.92) *[237]*. The numbers of exposures to the other ACTs (excluding artesunate-pyronaridine) were too small to allow for a subgroup analysis *[237]*. There is currently no documented record of the use of artesunate-pyronaridine during the first trimester of pregnancy. Combined with the known better tolerability and effectiveness and the longer duration of post-treatment prophylaxis, artemether-lumefantrine clearly has a more favourable risk–benefit profile than quinine for treating uncomplicated falciparum malaria in the first trimester.

An analysis of all exposures to artemisinin-based treatment (ABT) in the first trimester of pregnancy as a means of addressing the concerns previously demonstrated in animal studies showed no differences between pregnancies exposed in the first trimester to artemisinin and those exposed to non-ABT in terms of the composite adverse pregnancy outcome (ABT=42/736 [5.7%] vs non-ABT=96/1074 [8.9%]; aHR: 0.71; 95% CI: 0.49–1.03). Analysis for adverse pregnancy outcomes against the individual parameters in the composite analysis, including miscarriage, stillbirth or congenital anomalies, also revealed no statistically significant difference. There was also no difference in the risk of these adverse pregnancy outcomes when exposures were restricted to the putative embryo-sensitive period. This meta-analysis had 8126 additional pregnancies with 60 additional artemisinin exposures in the first trimester compared to the review published in 2017 *[238]*. This analysis *[237]* strengthens previous findings of the 2017 review that the potential for artemisinin-based embryotoxicity observed in animal studies is not reflected in humans treated for malaria. The analysis also demonstrates how few pregnancy outcomes after ACT exposures in the first trimester of pregnancy can be documented.

The teratogenic effect of the artemisinin observed in animal studies was not apparent in any of the reviewed data on human exposure to ACTs in the first trimester of pregnancy. However, there are some reasons to exercise caution in drawing a definite conclusion on the safety of the artemisinin derivatives as a drug class. These reasons include: the possibility of immortal time bias, resulting in an inability to detect early fetal losses; potential bias in observational study designs with exposure to quinine as the main comparator; and current limited postnatal evaluation, e.g. for cardiovascular and other malformations *[239]*. In addition, most of the safety data are on artemether-lumefantrine. Although artemisinin derivatives are rapidly converted to dihydroartemisinin as their active metabolite, differences between the different derivatives, or differences caused by the combination with different partner drugs cannot be excluded.

In terms of the safety and tolerability of the currently recommended ACT partner antimalarials, the antifolate sulfadoxine-pyrimethamine is contraindicated during the first trimester of pregnancy, as it is known to have a potential teratogenic risk in humans at therapeutic doses *[240]*. There is currently no documented exposure to pyronaridine in the first trimester of pregnancy. Among 3428 pregnant women in the second or third trimester treated with an ACT for *P. falciparum* malaria (at any parasite density and regardless of symptoms), drug-related adverse events such as asthenia, poor appetite, dizziness, nausea and vomiting occurred significantly more frequently in the artesunate-mefloquine group (50.6%) and the artesunate-amodiaquine group (48.5%) than in the dihydroartemisinin-piperaquine group (20.6%) and the artemether-lumefantrine group (11.5%) ($p<0.001$ for comparison among the four groups) *[241]*.

There is a lack of documented exposures very early in gestation (gestational weeks 4–10), which is considered a critical period for teratogenic risk. Therefore, the potential for any given medicine, including quinine, to cause a specific teratogenic effect can only be reliably ascertained when it has been administered during this sensitive period (which is almost impossible to study, given how soon this critical window occurs after the last menstrual period) *[239]*.

Certainty of the evidence

Low

The GDG judged the overall certainty of the assessed evidence across the different outcomes to be low due to bias inherent in observational studies. It was difficult to generalize across all ACTs because of the limited number of pregnant women in the first trimester treated with ACTs other than artemether-lumefantrine who were included in the review.

Values and preferences

The GDG judged that there may be important uncertainty or variability in patient values and preferences with regard to choosing between artemether-lumefantrine, other ACTs and quinine-based therapies, and in how different cultures would value the outcomes being monitored, such as perceptions around early trimester pregnancy losses, low birthweight and anaemia. However, artemether-lumefantrine compared to quinine is likely to be a more attractive option because of its greater availability and the convenience of a shorter, better tolerated treatment. Policy-makers and implementers will obviously prefer simplified recommendations on using artemether-lumefantrine or other ACTs to treat pregnant women with uncomplicated *P. falciparum* malaria across all trimesters.

According to a systematic review of sociocultural factors *[242]*, malaria in pregnancy is interpreted in locally defined categories, despite the higher malaria risk associated with pregnancy. Local context and health workers' ideas and comments influence concerns about malaria in pregnancy interventions. Factors such as the understanding of antenatal care, health worker–client interactions, household decision-making, gender relations, cost and distance to health facilities affect pregnant women's access to these interventions and their health-seeking behaviour. It is difficult to ascertain whether any sociocultural factors would result in variability in the likely preference for artemether-lumefantrine or other ACTs over quinine treatment.

Resources

Research evidence
ACTs, of which artemether-lumefantrine is the most widely used, are the treatment of choice for uncomplicated falciparum malaria in nearly all malaria-endemic countries and are thus readily available. Conversely, the supply of quinine has become problematic because of the small proportion of the total population that receive this antimalarial treatment. Clindamycin, which is recommended in combination with quinine, is commonly unavailable and unaffordable in most endemic regions. In addition, the quinine + clindamycin regime is associated with a high pill burden, requiring between 56 and 70 tablets to be ingested over a seven-day period. In most country programmes, quinine monotherapy is thus currently recommended in the first trimester of pregnancy. However, in some countries in eastern and southern Africa, quinine is rarely available in public facilities, and many pregnant women in the first trimester are already being treated with artemether-lumefantrine, based on reports from the national malaria programmes.

Aligning the first-line treatment of uncomplicated *P. falciparum* malaria for the first trimester with that currently recommended for the second and third trimesters would simplify case management, service delivery, communications and supply chain management. Such an alignment was assessed as likely to result in large savings. However, research on formal cost analysis and cost estimates regarding the use of artemether-lumefantrine or other ACTs versus quinine in the first trimester of pregnancy are still lacking.

Summary
ACTs, of which artemether-lumefantrine is the most widely used, are the treatment of choice for uncomplicated falciparum malaria in nearly all malaria-endemic countries and are thus readily available. Conversely, the supply of quinine has become problematic because of the small proportion of the total population that receive this antimalarial treatment. Clindamycin, which is recommended in combination with quinine, is commonly unavailable and unaffordable in most endemic regions. In addition, the quinine + clindamycin regime is associated with a high pill burden, requiring between 56 and 70 tablets to be ingested over a seven-day period. In most country programmes, quinine monotherapy is thus currently recommended in the first trimester of pregnancy. However, in some countries in eastern and southern Africa, quinine is rarely available in public facilities, and many pregnant women in the first trimester are already being treated with artemether-lumefantrine, based on reports from the national malaria programmes.

Aligning the first-line treatment of uncomplicated *P. falciparum* malaria for the first trimester with that currently recommended for the second and third trimesters would simplify case management, service delivery, communications and supply chain management. Such an alignment was assessed as likely to result in large savings. However, research on formal cost analysis and cost estimates regarding the use of artemether-lumefantrine or other ACTs versus quinine in the first trimester of pregnancy are still lacking.

Equity	Despite the obvious efficiency to be gained by harmonizing the treatment regimens throughout pregnancy, no studies were found. However, health equity will increase, especially for vulnerable populations, if this more effective, more accessible and better tolerated treatment is recommended for the management of malaria in all trimesters of pregnancy.
Acceptability	In considering the acceptability of artemether-lumefantrine versus quinine treatments, the GDG looked to how quinine is presently being used and accepted. Adherence to quinine is low because it is frequently associated with adverse effects, including cinchonism, nausea and hypoglycaemia *[231][243][244]*. In a review of 35 national guidelines, 66% recommended oral quinine as first-line treatment for uncomplicated malaria in the first trimester of pregnancy. Of these, only 29% included the combined use with clindamycin in their guidelines, reflecting the unavailability and/or cost of clindamycin *[245]*. Health care reliance on clinical diagnosis and poor adherence to treatment policy, especially in the first trimester, have been consistently reported. Prescribing practices have been driven by concerns over side effects and drug safety, patient preferences, drug availability and cost *[246]*. With poor adherence to the presently recommended quinine-based treatments and better access to ACTs, it appears that artemether-lumefantrine will be a more acceptable option. A three-day ACT treatment regimen is likely to be more acceptable than a seven-day treatment with quinine. Policy-makers and health care workers will likely welcome the evidence-based decision recommending artemether-lumefantrine for all trimesters of pregnancy. In situations where artemether-lumefantrine is no longer recommended for the treatment of malaria because of reduced efficacy and/or it is not promptly available, the use of some of the other ACTs recommended in national guidelines can be considered, given the demonstrated poorer outcomes of quinine treatment, along with the challenges of adherence to a seven-day treatment course. However, artesunate plus sulfadoxine-pyrimethamine cannot be recommended in the first trimester of pregnancy given the potential teratogenicity of antifolates. Furthermore, the lack of documented outcomes following the use of artesunate-pyronaridine precludes its use in the first trimester.
Feasibility	One consideration in determining the feasibility of the recommendation on treatment of malaria in the first trimester is that the existing warning against the use of artemisinin in the first trimester implies the need to consistently screen for pregnancy among all women of childbearing potential prior to treatment for malaria. However, pregnancy screening is rarely done prior to initiating malaria treatment. As observed by national programmes, the contraindication of artemisinin in the first trimester has resulted in confusion, most problematically resulting in pregnant women in the first trimester with severe malaria not receiving the recommended parenteral artesunate, thereby increasing malaria morbidity and mortality in this particularly vulnerable subgroup *[248][249]*. Given that ACTs, particularly artemether-lumefantrine, are already widely used in the treatment of malaria in pregnancy, although mainly in the second and third trimesters, uptake of artemether-lumefantrine (and other ACTs) should be feasible in the first trimester of pregnancy. There will also be less confusion once the recommendations are aligned across all trimesters of pregnancy, implying that artemether-lumefantrine or other ACTs should be more feasible and adherence to the implementation strategies should improve relative to that with quinine-based treatment.

Justification

The GDG reached a consensus on a strong recommendation for artemether-lumefantrine as the preferred treatment of uncomplicated *Plasmodium falciparum* malaria during the first trimester of pregnancy, despite the low certainty of evidence because:

- there was a large magnitude of beneficial effect of treatment on efficacy (demonstrated in the second and third trimesters of pregnancy), specifically a six-fold reduction in treatment failures following artemether-lumefantrine, compared to the currently recommended quinine-based therapies;
- artemether-lumefantrine was associated with trivial adverse events and significantly lower risk for adverse pregnancy

outcomes in the first trimester of pregnancy;
- artemether-lumefantrine had much better tolerability compared to quinine-based therapies; and
- there is probably increased equity, acceptability and feasibility, resulting from better access to artemether-lumefantrine and more efficient implementation of ACTs compared to quinine-based treatments.

Despite limited exposures to other ACTs (artesunate-amodiaquine, artesunate-mefloquine and dihydroartemisinin-piperaquine), the current evidence does not raise any concerns. However, consistent with the previous WHO recommendation that provided for limited use of ACTs if the first-line recommended medicine was not available, these other ACTs may be used where artemether-lumefantrine is not a recommended ACT for uncomplicated malaria or is not available, given the demonstrated poorer outcomes of quinine treatment, along with the challenges of adherence to a seven-day course of treatment. Exceptions are where the ACT partner drug is contraindicated, for example sulfadoxine-pyrimethamine, or its safety is unknown, for example pyronaridine. These three alternative ACTs (artesunate-amodiaquine, artesunate-mefloquine and dihydroartemisinin-piperaquine) are considered preferable to quinine-based treatments in the first trimester, as the latter are not as effective, not well tolerated and adherence is more challenging. Furthermore, quinine, the current WHO-recommended treatment, is associated with similar risks of poor birth outcomes compared to ACTs overall.

Research needs

- Although the safety of ACTs is reassuring and the independent patient data meta-analysis indicated that there is no apparent effect of gestational age on the risk of PCR-corrected treatment efficacy, collecting further evidence from clinical studies and close surveillance on the safety of ACT treatment in the first trimester must be continued and funded. This is particularly the case for artesunate-amodiaquine, artesunate-mefloquine, dihydroartemisinin-piperaquine and artesunate-pyronaridine, given the relatively few pregnant women in the first trimester with documented exposures to these drugs compared to artemether-lumefantrine.
- Operational studies including pregnancy registries are needed to strengthen pharmacovigilance among pregnant women and capture data from mass drug administration programmes and other interventions that may result in inadvertent exposures to ACTs during the first trimester.
- Continued pharmacovigilance and clinical research, such as controlled experimental studies on the efficacy and safety of antimalarial medicines, including new antimalarials, for the treatment of malaria in pregnancy, should be supported and funded given the high burden of malaria in pregnancy globally.

5.2.1.4.2 Young children and infants

Artemisinin derivatives are safe and well tolerated by young children; therefore, the choice of ACT is determined largely by the safety and tolerability of the partner drug.

SP (with artesunate) should be avoided in the first weeks of life because it displaces bilirubin competitively and could thus aggravate neonatal hyperbilibinaemia. Primaquine should be avoided in the first 6 months of life (although there are no data on its toxicity in infants), and tetracyclines should be avoided throughout infancy. With these exceptions, none of the other currently recommended antimalarial treatments has shown serious toxicity in infancy.

Delay in treating *P. falciparum* malaria in infants and young children can have fatal consequences, particularly for more severe infections. The uncertainties noted above should not delay treatment with the most effective drugs available. In treating young children, it is important to ensure accurate dosing and retention of the administered dose, as infants are more likely to vomit or regurgitate antimalarial treatment than older children or adults. Taste, volume, consistency and gastrointestinal tolerability are important determinants of whether the child retains the treatment. Mothers often need advice on techniques of drug administration and the importance of administering the drug again if it is regurgitated within 1 h of administration. Because deterioration in infants can be rapid, the threshold for use of parenteral treatment should be much lower.

Optimal antimalarial dosing in young children
Although dosing on the basis of body area is recommended for many drugs in young children, for the sake of simplicity, antimalarial drugs have been administered as a standard dose per kg bw for all patients, including young children and infants. This approach does not take into account changes in drug disposition that occur with development. The currently recommended doses of lumefantrine, piperaquine, SP, artesunate and chloroquine result in lower drug concentrations in young children and infants than in older patients. Adjustments to previous dosing regimens for dihydroartemisinin + piperaquine in uncomplicated malaria and for artesunate in severe malaria are now recommended to improve the drug

exposure in this vulnerable population. The available evidence for artemether + lumefantrine, SP and chloroquine does not indicate dose modification at this time, but young children should be closely monitored, as reduced drug exposure may increase the risk for treatment failure. Limited studies of amodiaquine and mefloquine showed no significant effect of age on plasma concentration profiles.

In community situations where parenteral treatment is needed but cannot be given, such as for infants and young children who vomit antimalarial drugs repeatedly or are too weak to swallow or are very ill, give rectal artesunate and transfer the patient to a facility in which parenteral treatment is possible. Rectal administration of a single dose of artesunate as pre-referral treatment reduces the risks for death and neurological disability, as long as this initial treatment is followed by appropriate parenteral antimalarial treatment in hospital. Further evidence on pre-referral rectal administration of artesunate and other antimalarial drugs is given in section 5.5.3 Treating severe malaria - pre-referral treatment options.

Optimal antimalarial dosing in infants
See recommendation for Infants less than 5 kg body weight below.

Optimal antimalarial dosing in malnourished young children
Malaria and malnutrition frequently coexist. Malnutrition may result in inaccurate dosing when doses are based on age (a dose may be too high for an infant with a low weight for age) or on weight (a dose may be too low for an infant with a low weight for age). Although many studies of the efficacy of antimalarial drugs have been conducted in populations and settings where malnutrition was prevalent, there are few studies of the disposition of the drugs specifically in malnourished individuals, and these seldom distinguished between acute and chronic malnutrition. Oral absorption of drugs may be reduced if there is diarrhoea or vomiting, or rapid gut transit or atrophy of the small bowel mucosa. Absorption of intramuscular and possibly intrarectal drugs may be slower, and diminished muscle mass may make it difficult to administer repeated intramuscular injections to malnourished patients. The volume of distribution of some drugs may be larger and the plasma concentrations lower. Hypoalbuminaemia may reduce protein binding and increase metabolic clearance, but concomitant hepatic dysfunction may reduce the metabolism of some drugs; the net result is uncertain.

Small studies of the pharmacokinetics of quinine and chloroquine showed alterations in people with different degrees of malnutrition. Studies of SP in IPTp and of amodiaquine monotherapy and dihydroartemisinin + piperaquine for treatment suggest reduced efficacy in malnourished children. A pooled analysis of data for individual patients showed that the concentrations of lumefantrine on day 7 were lower in children < 3 years who were underweight for age than in adequately nourished children and adults. Although these findings are concerning, they are insufficient to warrant dose modifications (in mg/kg bw) of any antimalarial drug in patients with malnutrition.

Strong recommendation for

Young children and infants (2015)

Infants weighing < 5 kg with uncomplicated *P. falciparum* malaria should be treated with an ACT at the same mg/kg bw target dose as for children weighing 5 kg.

*Not evaluated using the GRADE framework

Practical info

The pharmacokinetics properties of many medicines in infants differ markedly from those in adults because of the physiological changes that occur in the first year of life. Accurate dosing is particularly important for infants. The only antimalarial agent that is currently contraindicated for infants (< 6 months) is primaquine.

ACT is recommended and should be given according to body weight at the same mg/kg bw dose for all infants, including those weighing < 5 kg, with close monitoring of treatment response. The lack of infant formulations of most antimalarial drugs often necessitates division of adult tablets, which can lead to inaccurate dosing. When available, paediatric formulations and strengths are preferred, as they improve the effectiveness and accuracy of ACT dosing.

Evidence to decision

Benefits and harms	Undesirable effects:

- There is some evidence that artemether + lumefantrine and dihydroartemisinin + piperaquine may achieve lower plasma concentrations in infants than in older children and adults.

Justification

Evidence supporting the recommendation
Data available were not suitable for evaluation using the GRADE methodology.

In most clinical studies, subgroups of infants and older children were not distinguished, and the evidence for young infants (< 5 kg) is insufficient for confidence in current treatment recommendations. Nevertheless, despite these uncertainties, infants need prompt, effective treatment of malaria. There is limited evidence that artemether + lumefantrine and dihydroartemisinin + piperaquine achieve lower plasma concentrations in infants than in older children and adults.

Other considerations
The Guideline Development Group considered the currently available evidence too limited to warrant formal evidence review at this stage, and was unable to recommend any changes beyond the status quo. Further research is warranted.

Rationale for the recommendation
Treat infants weighing < 5 kg with uncomplicated *P. falciparum* malaria with an ACT. The weight-adjusted dose should achieve the same mg/kg bw target dose as for children weighing 5 kg.

5.2.1.4.3 Patients co-infected with HIV

There is considerable geographical overlap between malaria and HIV infection, and many people are co-infected. Worsening HIV-related immunosuppression may lead to more severe manifestations of malaria. In HIV-infected pregnant women, the adverse effects of placental malaria on birth weight are increased. In areas of stable endemic malaria, HIV-infected patients who are partially immune to malaria may have more frequent, higher-density infections, while in areas of unstable transmission, HIV infection is associated with increased risks for severe malaria and malaria-related deaths. Limited information is available on how HIV infection modifies therapeutic responses to ACTs. Early studies suggested that increasing HIV-related immunosuppression was associated with decreased treatment response to antimalarial drugs. There is presently insufficient information to modify the general malaria treatment recommendations for patients with HIV/AIDS.

Patients co-infected with tuberculosis
Rifamycins, in particular rifampicin, are potent CYP3A4 inducers with weak antimalarial activity. Concomitant administration of rifampicin during quinine treatment of adults with malaria was associated with a significant decrease in exposure to quinine and a five-fold higher recrudescence rate. Similarly, concomitant rifampicin with mefloquine in healthy adults was associated with a three-fold decrease in exposure to mefloquine. In adults co-infected with HIV and tuberculosis who were being treated with rifampicin, administration of artemether + lumefantrine resulted in significantly lower exposure to artemether, dihydroartemisinin and lumefantrine (nine-, six- and three-fold decreases, respectively).There is insufficient evidence at this time to change the current mg/kg bw dosing recommendations; however, as these patients are at higher risk of recrudescent infections they should be monitored closely.

Practice Statement

Patients co-infected with HIV (2015)

In people who have HIV/AIDS and uncomplicated *P. falciparum* malaria, artesunate + SP is not recommended if they are being treated with co-trimoxazole, and artesunate + amodiaquine is not recommended if they are being treated with efavirenz or zidovudine.

Justification

More data are available on use of artemether + lumefantrine with antiretroviral treatment. A study in children with uncomplicated malaria in a high-transmission area of Africa showed a decreased risk for recurrent malaria after treatment with artemether + lumefantrine in children receiving lopinavir–ritonavir-based antiretroviral treatment as compared with non-nucleoside reverse transcriptase inhibitor-based antiretroviral treatment. Evaluation of pharmacokinetics in these children and in healthy volunteers showed significantly higher exposure to lumefantrine and lower exposure to dihydroartemisinin with lopinavir–ritonavir-based antiretroviral treatment, but no adverse consequences. Conversely, efavirenz-based antiretroviral treatment was associated with a two- to fourfold decrease in exposure to lumefantrine in healthy volunteers and malaria-infected adults and children, with increased rates of recurrent malaria after treatment. Close monitoring is required. Increasing artemether + lumefantrine dosing with efavirenz-based antiretroviral treatment has not yet been studied. Exposure to lumefantrine and other non-nucleoside reverse transcriptase inhibitor-based antiretroviral treatment, namely nevirapine and etravirine, did not show consistent changes that would require dose adjustment.

Studies of administration of quinine with lopinavir–ritonavir or ritonavir alone in healthy volunteers gave conflicting results. The combined data are insufficient to justify dose adjustment. Single-dose atovaquone–proguanil with efavirenz, lopinavir–ritonavir or atazanavir–ritonavir were all associated with a significantly decreased area under the concentration–time curve for atovaquone (two- to fourfold) and proguanil (twofold), which could well compromise treatment or prophylactic efficacy. There is insufficient evidence to change the current mg/kg bw dosing recommendations; however, these patients should also be monitored closely.

5.2.1.4.4 Non-immune travellers

Travellers who acquire malaria are often non-immune people living in cities in endemic countries with little or no transmission or are visitors from non-endemic countries travelling to areas with malaria transmission. Both are at higher risk for severe malaria. In a malaria-endemic country, they should be treated according to national policy, provided the treatment recommended has a recent proven cure rate > 90%. Travellers who return to a non-endemic country and then develop malaria present a particular problem, and the case fatality rate is often high; doctors in non-malarious areas may be unfamiliar with malaria and the diagnosis is commonly delayed, and effective antimalarial drugs may not be registered or may be unavailable. However, prevention of transmission or the emergence of resistance are not relevant outside malaria-endemic areas. If the patient has taken chemoprophylaxis, the same medicine should not be used for treatment. Treatment of *P. vivax*, *P. ovale* and *P. malariae* malaria in travellers should be the same as for patients in endemic areas (see section 5.4).

There may be delays in obtaining artesunate, artemether or quinine for the management of severe malaria outside endemic areas. If only parenteral quinidine is available, it should be given, with careful clinical and electrocardiographic monitoring (see section 5.5 Treating severe malaria).

Strong recommendation for , High certainty evidence

Non-immune travellers (2015)

Travellers with uncomplicated *P. falciparum* malaria returning to non-endemic settings should be treated with an ACT.

Evidence to decision

Certainty of the evidence High

Justification

GRADE
Studies have consistently demonstrated that the five WHO recommended ACTs have less than 5% PCR-adjusted treatment failure rates in settings without resistance to the partner drug (high quality evidence).

Other considerations
The Guideline Development Group considered the evidence of superiority of ACTs over non-ACTs from endemic settings to be equally applicable to those travelling from non-endemic settings.

5.2.1.4.5 Uncomplicated hyperparasitaemia

Uncomplicated hyperparasitaemia is present in patients who have ≥ 4% parasitaemia but no signs of severity. They are at increased risk for severe malaria and for treatment failure and are considered an important source of antimalarial drug resistance.

Practice Statement

Hyperparasitaemia (2015)

People with *P. falciparum* hyperparasitaemia are at increased risk for treatment failure, severe malaria and death and should be closely monitored, in addition to receiving an ACT.

Justification

In falciparum malaria, the risk for progression to severe malaria with vital organ dysfunction increases at higher parasite densities. In low-transmission settings, mortality begins to increase when the parasite density exceeds 100 000/µL (~2% parasitaemia). On the north-west border of Thailand, before the general introduction of ACT, parasitaemia > 4% without signs of severity was associated with a 3% mortality rate (about 30-times higher than from uncomplicated falciparum malaria with lower densities) and a six-times higher risk of treatment failure. The relationship between parasitaemia and risks depends on the epidemiological context: in higher-transmission settings, the risk of developing severe malaria in patients with high parasitaemia is lower, but "uncomplicated hyperparasitaemia" is still associated with a significantly higher rate of treatment failure.

Patients with a parasitaemia of 4–10% and no signs of severity also require close monitoring, and, if feasible, admission to hospital. They have high rates of treatment failure. Non-immune people such as travellers and individuals in low-transmission settings with a parasitaemia > 2% are at increased risk and also require close attention. Parasitaemia > 10% is considered to indicate severe malaria in all settings.

It is difficult to make a general recommendation about treatment of uncomplicated hyperparasitaemia, for several reasons: recognizing these patients requires an accurate, quantitative parasite count (they will not be identified from semi-quantitative thick film counts or RDTs), the risks for severe malaria vary considerably, and the risks for treatment failure also vary. Furthermore, little information is available on therapeutic responses in uncomplicated hyperparasitaemia. As the artemisinin component of an ACT is essential in preventing progression to severe malaria, absorption of the first dose must be ensured (atovaquone – proguanil alone should not be used for travellers presenting with uncomplicated hyperparasitaemia). Longer courses of treatment are more effective; both giving longer courses of ACT and preceding the standard 3-day ACT regimen with parenteral or oral artesunate have been used.

5.2.1.5 Uncomplicated malaria caused by P. vivax, P. ovale, P. malariae or P. knowlesi

Plasmodium vivax accounts for approximately half of all malaria cases outside Africa *[3][250][251]*. It is prevalent in the Middle East, Asia, the Western Pacific and Central and South America. With the exception of the Horn, it is rarer in Africa, where there is a high prevalence of the Duffy-negative phenotype, particularly in West Africa, although cases are reported in both Mauritania and Mali *[251]*. In most areas where *P. vivax* is prevalent, the malaria transmission rates are low (except on the island of New Guinea). Affected populations achieve only partial immunity to this parasite, and so people of all ages are at risk for *P. vivax* malaria *[251]*. Where both *P. falciparum* and *P. vivax* are prevalent, the incidence rates of *P. vivax* tend to peak at a younger age than for *P. falciparum*. This is because each *P. vivax* inoculation may be followed by several relapses. The other human malaria parasite species, *P. malariae* and *P. ovale* (which is in fact two sympatric species), are less common. *P. knowlesi*, a simian parasite, causes occasional cases of malaria in or near forested areas of South-East Asia and the Indian subcontinent *[252]*. In parts of the island of Borneo, *P. knowlesi* is the predominant cause of human malaria and an important cause of severe malaria

Of the six species of *Plasmodium* that affect humans, only *P. vivax* and the two species of *P. ovale [253]* form hypnozoites, which are dormant parasite stages in the liver that cause relapse weeks to years after the primary infection. *P. vivax* preferentially invades reticulocytes, and repeated illness causes chronic anaemia, which can be debilitating and sometimes life-threatening, particularly in young children *[254]*. Recurrent vivax malaria is an important impediment to human and economic development in affected populations. In areas where *P. falciparum* and *P. vivax* co-exist, intensive malaria control often has a greater effect on *P. falciparum*, as *P. vivax,* is more resilient to interventions.

Although *P. vivax* has been considered to be a benign form of malaria, it may sometimes cause severe disease *[255]*. The major complication is anaemia in young children. In Papua province, Indonesia *[255]*, and in Papua New Guinea *[256]*, where malaria transmission is intense, *P. vivax* is an important cause of malaria morbidity and mortality, particularly in young infants and children. Occasionally, older patients develop vital organ involvement similar to that in severe and complicated *P. falciparum* malaria *[257][258]*. During pregnancy, infection with *P. vivax*, as with *P. falciparum*, increases the risk for abortion and reduces birth weight *[259][207]*. In primigravidae, the reduction in birth weight is approximately two thirds that associated with *P. falciparum.* In one large series, this effect increased with successive pregnancies *[259]*.

P. knowlesi is a zoonosis that normally affects long- and pig-tailed macaque monkeys. It has a daily asexual cycle, resulting in a rapid replication rate and high parasitaemia. *P. knowlesi* may cause a fulminant disease similar to severe falciparum malaria (with the exception of coma, which does not occur) *[260][261]*. Co-infection with other species is common.

Diagnosis
Diagnosis of *P. vivax*, *P. ovale*, and *P. malariae* malaria is based on microscopy. *P. knowlesi* is frequently misdiagnosed under the microscope, as the young ring forms are similar to those of *P. falciparum*, the late trophozoites are similar to those of *P. malariae*, and parasite development is asynchronous. Rapid diagnostic tests based on immunochromatographic methods are available for the detection of *P. vivax* malaria; however, they are relatively insensitive for detecting *P. malariae* and *P. ovale* parasitaemia. Rapid diagnostic antigen tests for human *Plasmodium* species show poor sensitivity for *P. knowlesi* infections in humans with low parasitaemia *[262]*.

Treatment
The objectives of treatment of vivax malaria are twofold: to cure the acute blood stage infection and to clear hypnozoites from the liver to prevent future relapses. This is known as “radical cure”.

In areas with chloroquine-sensitive *P. vivax*
For chloroquine-sensitive vivax malaria, oral chloroquine at a total dose of 25 mg base/kg bw is effective and well tolerated. Lower total doses are not recommended, as these encourage the emergence of resistance. Chloroquine is given at an initial dose of 10 mg base/kg bw, followed by 10 mg/kg bw on the second day and 5 mg/kg bw on the third day. In the past, the initial 10 mg/kg bw dose was followed by 5 mg/kg bw at 6 h, 24 h and 48 h. As residual chloroquine suppresses the first relapse of tropical *P. vivax* (which emerges about 3 weeks after onset of the primary illness), relapses begin to occur 5–7 weeks after treatment if radical curative treatment with primaquine is not given.

ACTs are highly effective in the treatment of vivax malaria, allowing simplification (unification) of malaria treatment; i.e. all malaria infections can be treated with an ACT. The exception is artesunate + SP, where resistance significantly compromises its efficacy. Although good efficacy of artesunate + SP was reported in one study in Afghanistan, in several other areas (such as South-East Asia) *P. vivax* has become resistant to SP more rapidly than *P. falciparum.* The initial response to all ACTs is rapid in vivax malaria, reflecting the high sensitivity to artemisinin derivatives, but, unless primaquine is given, relapses commonly follow. The subsequent recurrence patterns differ, reflecting the elimination kinetics of the partner drugs. Thus, recurrences, presumed to be relapses, occur earlier after artemether + lumefantrine than after dihydroartemisinin + piperaquine or artesunate + mefloquine because lumefantrine is eliminated more rapidly than either mefloquine or piperaquine. A similar temporal pattern of recurrence with each of the drugs is seen in the *P. vivax* infections that follow up to one third of acute falciparum malaria infections in South-East Asia.

In areas with chloroquine-resistant *P. vivax*
ACTs containing piperaquine, mefloquine or lumefantrine are the recommended treatment, although artesunate + amodiaquine may also be effective in some areas.

In the systematic review of ACTs for treating *P. vivax* malaria, dihydroartemisinin + piperaquine provided a longer prophylactic effect than ACTs with shorter half-lives (artemether + lumefantrine, artesunate + amodiaquine), with significantly fewer recurrent parasitaemias during 9 weeks of follow-up (RR, 0.57; 95% CI, 0.40–0.82, three trials, 1066 participants). The half-life of mefloquine is similar to that of piperaquine, but use of dihydroartemisinin + piperaquine in *P. vivax* mono-infections has not been compared directly in trials with use of artesunate + mefloquine.

Uncomplicated *P. ovale*, *P. malariae* or *P. knowlesi* malaria
Resistance of *P. ovale, P. malariae* and *P. knowlesi* to antimalarial drugs is not well characterized, and infections caused by these three species are generally considered to be sensitive to chloroquine. In only one study, conducted in Indonesia, was resistance to chloroquine reported in *P. malariae*.

The blood stages of *P. ovale*, *P. malariae* and *P. knowlesi* should therefore be treated with the standard regimen of ACT or chloroquine, as for vivax malaria.

Mixed malaria infections
Mixed malaria infections are common in endemic areas. For example, in Thailand, despite low levels of malaria transmission, 8% of patients with acute vivax malaria also have *P. falciparum* infections, and one third of acute *P. falciparum* infections are followed by a presumed relapse of vivax malaria (making vivax malaria the most common complication of falciparum malaria).

Mixed infections are best detected by nucleic acid-based amplification techniques, such as PCR; they may be underestimated with routine microscopy. Cryptic *P. falciparum* infections in vivax malaria can be revealed in approximately 75% of cases by RDTs based on the *Pf*HRP2 antigen, but several RDTs cannot detect mixed infection or have low sensitivity for detecting cryptic vivax malaria. ACTs are effective against all malaria species and so are the treatment of choice for mixed infections.

Practice Statement

Blood stage infection (2015)

If the malaria species is not known with certainty, adults and children should be treated as for uncomplicated *P. falciparum* malaria.

Strong recommendation for , High certainty evidence

Blood stage infection (2015)

In areas with chloroquine-susceptible infections, adults and children with uncomplicated *P. vivax, P. ovale, P. malariae* or *P. knowlesi* malaria should be treated with either an ACT or chloroquine.

In areas with chloroquine-resistant infections, adults and children with uncomplicated *P. vivax, P. ovale, P. malariae* or *P. knowlesi* malaria should be treated with an ACT.

Practical info

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For chloroquine-sensitive vivax malaria, oral chloroquine at a total dose of 25 mg base/kg bw is effective and well tolerated. Lower total doses are not recommended, as these encourage the emergence of resistance. Chloroquine is given at an initial dose of 10 mg base/kg bw, followed by 10 mg/kg bw on the second day and 5 mg/kg bw on the third day. In the past, the initial 10-mg/kg bw dose was followed by 5 mg/kg bw at 6 h, 24 h and 48 h. As residual chloroquine suppresses the first relapse of tropical *P. vivax* (which emerges about 3 weeks after onset of the primary illness), relapses begin to occur 5–7 weeks after treatment if radical curative treatment with primaquine is not given.

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Uncomplicated *P. ovale, P. malariae* or *P. knowlesi* malaria
Resistance of *P. ovale, P. malariae* and *P. knowlesi* to antimalarial drugs is not well characterized, and infections caused by these three species are generally considered to be sensitive to chloroquine. In only one study, conducted in Indonesia, was resistance to chloroquine reported in *P. malariae*.

The blood stages of *P. ovale, P. malariae* and *P. knowlesi* should therefore be treated with the standard regimen of ACT or chloroquine, as for vivax malaria.

Mixed Malaria Infections
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Mixed infections are best detected by nucleic acid-based amplification techniques, such as PCR; they may be underestimated with routine microscopy. Cryptic *P. falciparum* infections in vivax malaria can be revealed in approximately 75% of cases by RDTs based on the PfHRP2 antigen, but several RDTs cannot detect mixed infection or have low sensitivity for detecting cryptic vivax malaria. ACTs are effective against all malaria species and so are the treatment of choice for mixed infections.

Evidence to decision

Benefits and harms

Desirable effects:

- ACTs clear parasites more quickly than chloroquine (high-quality evidence).
- ACTs with long half-lives provide a longer period of suppressive post-treatment prophylaxis against relapses and new infections (high-quality evidence).
- Simplified national protocols for all forms of uncomplicated malaria.
- Adequate treatment of undiagnosed *P. falciparum* in mixed infections.

Certainty of the evidence

High

Overall certainty of evidence for all critical outcomes: high.

Justification

GRADE
In a systematic review of ACTs for the treatment of *P. vivax* malaria *[252]*, five trials were conducted in Afghanistan, Cambodia, India, Indonesia and Thailand between 2002 and 2011 with a total of 1622 participants which compared ACTs directly with chloroquine. In comparison with chloroquine:

ACTs cleared parasites from the peripheral blood more quickly (parasitaemia after 24 h of treatment: RR, 0.42; 95% CI, 0.36–0.50, four trials, 1652 participants, high-quality evidence); and

ACTs were at least as effective in preventing recurrent parasitaemia before day 28 (RR, 0.58; 95% CI, 0.18–1.90, five trials, 1622 participants, high-quality evidence).

In four of these trials, few cases of recurrent parasitaemia were seen before day 28 with both chloroquine and ACTs. In the fifth trial, in Thailand in 2011, increased recurrent parasitaemia was seen after treatment with chloroquine (9%), but was infrequent after ACT (2%) (RR, 0.25; 95% CI, 0.09–0.66, one trial, 437 participants).

ACT combinations with long half-lives provided a longer prophylactic effect after treatment, with significantly fewer cases of recurrent parasitaemia between day 28 and day 42 or day 63 (RR, 0.57; 95% CI, 0.40–0.82, three trials, 1066 participants, moderate-quality evidence).

Other considerations
The guideline development group recognized that, in the few settings in which *P. vivax* is the only endemic species and where chloroquine resistance remains low, the increased cost of ACT may not be worth the small additional benefits. Countries where chloroquine is used for treatment of vivax malaria should monitor for chloroquine resistance and change to ACT when the treatment failure rate is > 10% at day 28.

Remarks
Current methods cannot distinguish recrudescence from relapse or relapse from newly acquired infections, but the aim of treatment is to ensure that the rates of recurrent parasitaemia of any origin are < 10%.

Primaquine has significant asexual stage activity against vivax malaria and augments the therapeutic response to chloroquine. When primaquine is given routinely for 14 days, it may mask low-level chloroquine resistance and prevent vivax recurrence within 28 days.

Rationale for the recommendation
The Guideline Development Group recognized that, in the few settings in which *P. vivax* is the only endemic species and where chloroquine resistance remains low, the increased cost of ACT may not be worth the small additional benefits. In these settings, chloroquine may still be considered, but countries should monitor chloroquine resistance and change to ACT when the treatment failure rate is > 10% on day 28.

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Remarks

Current methods do not distinguish recrudescence from relapse or relapse from newly acquired infection, but the aim of treatment is to ensure that the rates of recurrent parasitaemia of any origin is < 10% within 28 days.

When primaquine is not given for radical cure, slowly eliminated ACT that prevents recurrent parasitaemia before day 28 should be used (dihydroartemisinin + piperaquine or artesunate + mefloquine).

Primaquine has significant asexual stage activity against vivax malaria and augments the therapeutic response to chloroquine. When primaquine is given routinely for 14 days, it may mask low-level chloroquine resistance and prevent vivax recurrence within 28 days.

When primaquine is given routinely for 14 days, ACTs with shorter half-lives (artemether + lumefantrine, or artesunate + amodiaquine) may be sufficient to keep the rate of recurrent parasitaemia before day 28 below 10%.

Rationale for the recommendation
The Guideline Development Group recognized that, in the few settings in which *P. vivax* is the only endemic species and where chloroquine resistance remains low, the increased cost of ACT may not be worth the small additional benefits. In these settings, chloroquine may still be considered, but countries should monitor chloroquine resistance and change to ACT when the treatment failure rate is > 10% on day 28.

5.2.1.6 Testing for glucose-6-phosphate dehydrogenase (G6PD) deficiency

Testing for glucose-6-phosphate-dehydrogenase (G6PD) deficiency should be done to inform administration of the appropriate treatment to prevent relapses of *P. vivax* and *P. ovale*. Since G6PD deficiency is a genetic abnormality, if a person is already aware of their G6PD status through G6PD spectrophotometry, no additional test may be required. However, the majority of persons have not been tested before, and therefore a near-patient test may have to be used. In the context of these guidelines, near-patient testing refers to testing performed close to the patient but not necessarily at the bedside. Near-patient testing may be more technically complex and/or need some basic infrastructure, such as electricity or a device, and for this the patient may need to be referred to a nearby testing facility, rather than having the test conducted immediately at the point of care.

There are presently two types of near-patient G6PD tests available – qualitative and semi-quantitative tests. Qualitative tests can discriminate between persons who have G6PD activity < 30% of normal G6PD activity (and are therefore classified as G6PD deficient), and persons who have G6PD activity > 30% of normal (and are classified as non-deficient). In fact, they are almost certainly non-deficient if male, and either non-deficient or with intermediate deficiency if females. Semi-quantitative tests, instead, can classify persons as G6PD normal (≥ 70% G6PD activity), or G6PD deficient (< 30% G6PD activity) or intermediate G6PD deficiency (between 30 and 70% of normal activity).

No head-to-head comparisons have been done between the accuracy and/or cost-effectiveness of qualitative near patients G6PD tests and semi-quantitative near-patient G6PD tests. The choice between the two depends on available resources, as the semi-quantitative test requires more expensive equipment and trained personnel. However, it must be noted that tafenoquine is contraindicated in patients with G6PD deficiency, if G6PD status is unknown, or if G6PD activity ≤ 70%.Therefore tafenoquine can only be used when the result of a quantitative or a semi-quantitative test is available. See Section 5.2.1.7 anti-relapse treatment of *P. vivax* and *P. ovale*.

Practice Statement

Blood stage infection (2024)

The G6PD status of patients should be used to guide administration of either primaquine or tafenoquine for preventing relapse.

Practical info

Please refer to *Testing for G6PD deficiency for safe use of primaquine in radical cure of P. vivax and P. ovale (Policy brief)* *[264]* and *Guide to G6PD deficiency rapid diagnostic testing to support P. vivax radical cure* *[265]*.

If G6PD testing is not available, a decision to prescribe or withhold primaquine should be based on the balance of the probability and benefits of preventing relapse against the risks of primaquine-induced haemolytic anaemia. This depends on the population prevalence of G6PD deficiency, the severity of the prevalent genotypes, the daily dose of primaquine and on the capacity of health services to identify and manage primaquine-induced haemolytic reactions. Tafenoquine should not be deployed without accurate G6PD quantitative or semi-quantitative testing and should only be used if patients have ≥ 70% G6PD activity. See Section 5.2.1.7 anti-relapse treatment of *P. vivax* and *P. ovale*.

Strong recommendation for , Moderate certainty evidence

Qualitative near-patient G6PD tests (2024)

Qualitative near-patient tests for G6PD deficiency should be used to inform administration of specific treatment regimens to prevent relapses of *P. vivax* and *P. ovale*. G6PD non-deficient individuals can receive 0.5 mg/kg/day primaquine for 14 days or 0.5 mg/kg/day primaquine for 7 days.

- *In males and females, <30% of normal G6PD activity is considered deficient.*
- *In patients undergoing G6PD activity testing, near-patient qualitative tests for G6PD deficiency are considered highly accurate to distinguish G6PD above or below a threshold of 30% of normal G6PD activity.*
- *These tests cannot be used to identify females with intermediate G6PD deficiency (30–70% G6PD activity) due to a heterozygous genotype. Instead, females with G6PD activity in this intermediate range will be classified as normal with a qualitative test.*

Evidence to decision

Benefits and harms	The main benefits are (1) the prevention of haemolysis in patients who are G6PD deficient, while at the same time providing them with effective treatment; and (2) higher confidence in prescribing the correct treatment by health care professionals. Prevention of haemolysis is the most important desirable effect of implementing G6PD testing, and this was judged to have a large effect based on the sensitivity and specificity of available G6PD tests. Possible harms include (1) potentially more relapses in a small group of patients as false positives will be prescribed a longer course of treatment (8 weekly doses of primaquine) associated with lower adherence to treatment compared to standard (daily doses of) treatment; (2) a false sense of security which causes less attention for potential haemolysis in false negative tested patients; and (3) after confirmation of *P. vivax* infection a second finger prick is needed to determine the G6PD status. These harms of implementing G6PD testing were considered to have a trivial to small effect based on the sensitivity and specificity of available G6PD tests.

Certainty of the evidence

Moderate

Diagnostic accuracy (sensitivity and specificity)
The test is considered to be very accurate, with a pooled sensitivity of 94.9% (95% CI 89.4, 97.6) and a pooled specificity of 96.2% (95% CI 93.5, 97.8). See Table “Summary sensitivity and sensitivity of the qualitative tests by threshold” under Evidence profiles/Research evidence.

As the summary estimates for sensitivity and specificity are clearly above the values stated in the target product profile (TPP) *[247]*, and as the lower boundaries of the confidence intervals are close to this limit for the sensitivity or above this limit for the specificity, the accuracy is considered to be very high.

Furthermore, given the low prevalence of individuals with G6PD activity below 30%, the absolute numbers of false negative test results will always be low to very low (<1% of all negatives).

Explanation
A sensitivity of 94.9% means that of all patients who are G6PD deficient (i.e. who have a G6PD activity of less than 30% of normal activity based on spectrophotometry), 94.9% will have a qualitative point-of-care test result indicating G6PD deficiency. The remaining 5.1% of the G6PD deficient patients will have a test result indicating that they are not deficient (false negative test result).

A specificity of 96.2% means that of all patients who have a G6PD activity above 30% (based on spectrophotometry), 96.2% will have a qualitative point-of-care test result indicating that they are NOT G6PD deficient. This does include patients with intermediate G6PD activity of whom the test indicates that they are not deficient. On the contrary, 3.8% of the patients with a normal or intermediate G6PD activity will have a test result indicating that they are deficient (false positive test result).

If a test with these characteristics would be applied to a group of 10,000 patients in which the 5% of patients are G6PD deficient (<30% of normal activity), then this would result in a total of 836 positive test results and thus 836 patients for whom a weekly regimen of primaquine during 8 weeks would be suggested, while 361 of them actually have a normal or intermediate G6PD activity (and 475 will indeed be G6PD deficient). This would also result then in 9164 negative test results and thus 9164 patients who will receive daily doses of 0.5 mg/kg/day primaquine for 14 days or 0.5 mg/kg/day primaquine for 7 days, while 25 of them would actually be G6PD deficient (and 9139 will not be deficient but some of them, especially females, may have intermediate activity).

Guidance
The practical guidance on the use of qualitative near-patient tests for G6PD deficiency should include all aspects of safe implementation of a new diagnostic test e.g. implementation plan, clear national guidelines, quality assurance and prequalification of tests, training of users, quality assurance of testing, and selection of the type of health services where these tests should be deployed. In addition the guidance should also include specific information on the anti-relapse primaquine treatment regimens linked with a negative (normal) test result, i.e. 0.5 mg/kg/day primaquine for 14 days or 0.5 mg/kg/day primaquine for 7 days, and with positive (deficient) test results, i.e. 0.75 mg/kg weekly doses of primaquine for 8 weeks.

Like for the introduction of any new diagnostic tests, this should be implemented with a sound quality assurance to address the specific requirements for the newly introduced diagnostics, reaching all levels of the health care system where these will be co-deployed with the anti-relapse *P. vivax* treatment.

Conclusion
We are moderately confident in the estimates of sensitivity and specificity; the true sensitivity and specificity are likely to be close to the estimates of the sensitivity and specificity mentioned here, but there is a possibility that it is substantially different. This is because: i) some of the studies on which the summary estimates were based, had a high risk of bias; and ii) the prevalence of G6PD deficiency was different in some study sites or some studies only included one gender (indirectness). See Table “Certainty of the evidence table for near-patients qualitative G6PD tests” under Evidence profiles/ Research evidence.

	We are less certain for other estimates of the evidence of effects on clinical management effects (the links between test results and management decisions), because of the absence of observational studies or availability of modelling studies only.
Values and preferences	Ministries of health ask for evidence regarding the added value of G6PD testing to inform their decision to implement G6PD testing to prevent haemolysis cases while still being able to treat patients for malaria and prevent relapse. For most outcomes, no or little variation in values may be expected, except maybe people with false positive results, considered G6PD deficient. These will receive the weekly primaquine treatment and may experience lower protection due to poor adherence compared to daily primaquine regimen, while they do not need protection against primaquine-induced haemolysis.
Resources	The resources required for G6PD qualitative tests are balanced by the benefits of testing. Qualitative tests have relatively low to moderate costs (less than US$ 5), but the cost of setting up diagnostic/ screening test programmes is significant (and separate from using a single test on an individual patient); this would also include indirect costs, such as training. Still, there would be savings because of fewer relapses and less hospitalisations.
Equity	The impact is likely to be increased equity, as qualitative tests treat males and females similarly. With respect to gender the qualitative test is less accurate than to the semi-quantitative test because it will correctly classify most males (deficient or non-deficient), but it will not separate females with intermediate deficiency from females who are non-deficient. The scoping review of contextual factors did express concerns regarding potential negative effects on equity in remote villages and regarding *P. vivax* malaria, but the committee judged that absence of testing may have larger negative impact on equity than testing (Barker et al., unpublished evidence).
Acceptability	The qualitative tests are considered to be probably acceptable to all stakeholders, according to the findings in the scoping review of contextual factors.
Feasibility	The implementation of qualitative test is considered to be feasible, although there may be settings where there may be barriers to the use of qualitative tests.

Justification

Although the certainty about the sensitivity of 94.9% (95% CI 89.4, 97.6) and the specificity of 96.2% (95% CI 93.5, 97.8) is moderate due to risk of bias and indirectness, the numbers of false negatives (and thus patients with a high risk of haemolysis) are very low: this would consist of 1 to 11 patients in a total of 1000 patients in a situation where the prevalence of G6PD deficiency is 5% or 10% in the whole population. The number of false positives is much higher, but the immediate consequences for this group are less severe. They will receive a longer treatment regimen, i.e. primaquine once weekly for eight weeks instead of once daily for 14 days, with potential risks of non-adherence.

However, without testing, the G6PD activity status of malaria patients remains unknown, and may lead to totally refraining from anti-relapse treatment. As this would lead to suboptimal treatment for malaria on a large scale (and many more relapses), we consider that there is a large positive effect of qualitative testing for G6PD deficiency (defined as distinguishing patients with <30% G6PD activity from those with >30 G6PD activity) on choosing the appropriate anti-relapse treatment.

Research needs

The scoping review on contextual factors showed that there is an evidence need around contextual factors, especially the evidence of the effects of G6PD testing implementation on clinical management and risks of haemolysis (links between test results and treatment decisions) and on equity and human rights issues (Barker *et al.*, unpublished evidence).

More evidence may be needed to provide a higher certainty of the diagnostic accuracy for all near-patients qualitative G6PD tests that are available on the market.

Strong recommendation for , Moderate certainty evidence

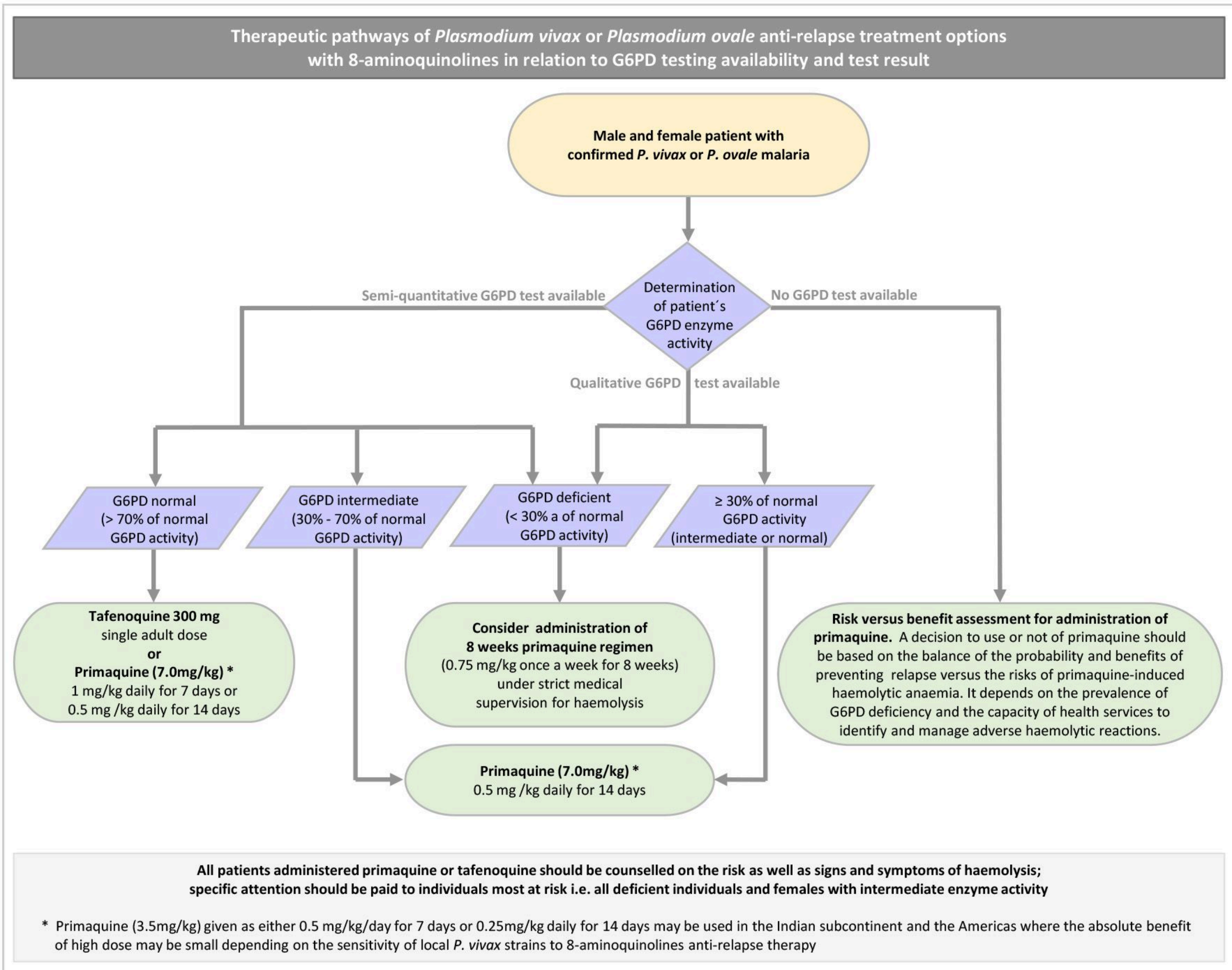
Semi-quantitative near-patient G6PD tests (2024)

Semi-quantitative near-patient tests with fixed standard thresholds for deficient, intermediate and normal G6PD activity should be used to inform administration of specific treatment regimens. The dose of 1 mg/kg/day primaquine for 7 days or single dose tafenoquine should only be given to those above the threshold that corresponds to >70% of normal G6PD activity; and 0.5 mg/kg/day primaquine for 14 days or 0.5 mg/kg/day primaquine for 7 days can be given to those with a threshold that corresponds to > 30% of normal G6PD activity to prevent relapses of *P. vivax* and *P. ovale*.

- *In males and females, <30% of normal G6PD activity is considered deficient; females with G6PD activity between 30% and 70% due to a heterozygous genotype are considered to have intermediate G6PD activity and are also (but less so) at risk of haemolysis.*
- *In patients undergoing G6PD activity testing, near-patient semi-quantitative tests for G6PD deficiency with fixed thresholds corresponding to >30% and <70% of normal G6PD activity are considered highly accurate at a threshold of 30% of normal G6PD activity to indicate whether P. vivax and P. ovale patients are G6PD deficient, and are considered accurate at a threshold of ≤ 70% activity to indicate whether P. vivax and P. ovale patients are deficient or have intermediate G6PD activity.*

Practical info

Therapeutic pathways of *P. vivax* and *P. ovale* anti-relapse treatment with 8-aminoquinolines in relation to G6PD testing



In order to prevent relapses of *P. vivax* and *P. ovale*, and when the G6PD status of the patient was previously unknown, the following recommendations are made:

A. If only a qualitative near-patient test for G6PD deficiency is available, tafenoquine single dose treatment or high dose primaquine (1mg/kg/day for 7 days) should not be given. If by the qualitative test the patient is classified as non-deficient primaquine should be used at a regimen of 0.5 mg/kg/day for 14 days or for 7 days. Since many heterozygous females will be classified as non-deficient by this type of test, precautions should be used (see Figure above). If the patient tests G6PD deficient, consider primaquine 0.75 mg/Kg once a week for 8 weeks under medical supervision and surveillance for haemolysis.

B. If a semi-quantitative near-patient G6PD test is available:

a. if the patient has G6PD activity > 70%, tafenoquine as single dose tafenoquine or high dose primaquine (1 mg/kg/day for 7 days or 0.5 mg/kg daily for 14 days) can be given;

b. if the patient has intermediate G6PD deficiency between 30 and 70% tafenoquine should not be given; primaquine should be used at a regimen of 0.5 mg/kg/day for 14 days or for 7 days, with precautions;

c. if the patient has G6PD activity < 30%, consider primaquine 0.75 mg/kg once a week for 8 weeks under medical supervision and surveillance for haemolysis.

C. If G6PD testing is not available, a decision to prescribe or withhold primaquine should be based on the balance of the probability and benefits of preventing relapse against the risks of primaquine-induced haemolytic anaemia. This depends on the population prevalence of G6PD deficiency, the severity of the prevalent genotypes, the daily dose of primaquine and on the capacity of health services to identify and manage primaquine-induced haemolytic reactions. Tafenoquine should not be deployed if accurate G6PD quantitative or semi-quantitative testing is not available.

Evidence to decision

Benefits and harms

The main benefits are (1) the prevention of haemolysis in patients who are G6PD deficient, while at the same time providing them with effective treatment; and (2) higher confidence in prescribing the correct treatment by health care professionals. Prevention of haemolysis is the most important desirable effect of implementing G6PD testing, and this benefit was judged to have a large effect, based on sensitivity and specificity data of available G6PD tests.

Possible harms include (1) potentially more relapses in a small group of patients as false positives will be prescribed a longer course of treatment (8 weekly doses of primaquine) compared to situations where all patients receive standard (single or daily doses of) treatment; (2) a false sense of security which causes less attention for potential haemolysis in false negative tested patients; and (3) following confirmation of *P. vivax* infection a second finger prick is needed to determine the G6PD status. These harms were considered to have a trivial to small effect, based on sensitivity and specificity data of available G6PD tests.

Certainty of the evidence

Moderate

Diagnostic accuracy (sensitivity and specificity)

The Standard G6PD test used [i] with STANDARD G6PD Analyzer (SD Biosensor, Inc) and manufacturer references to calculate relevant thresholds is considered to be very accurate at the 30% threshold, with a pooled sensitivity of 100% (95% CI 98.2, 100) and a summary specificity of 97.0% (95% CI 96.5, 97.5). This test is considered to be accurate when used at the 70% threshold, with a summary sensitivity of 91.4% (95% CI 75.5, 97.4) and a summary specificity of 93.7% (95% CI 85.8, 97.4). See Table “Standard G6PD by SD Biosensor – Manufacturer determined thresholds, as per the instructions for use” under Evidence profiles/Research evidence.

The summary estimates for sensitivity and specificity (and the lower boundaries of the confidence intervals for 30% threshold) are clearly above the values stated in the WHO target product profile (TPP) for G6PD tests *[247]*. Given the low prevalence of G6PD deficient patients, the absolute numbers of false negative test results will always be low to very low (<1% of all negatives). By pooled or individual results, no product included in the systematic review met the TPP requirements at all relevant thresholds when the adjusted male median (AMM) was used to define thresholds of G6PD activity using spectrophotometry (See Table “Summary sensitivity and sensitivity of the semiquantitative tests by threshold as defined by the adjusted male median (AMM)” under Evidence profiles/Research evidence). Therefore, the analysis was done using manufacturer defined thresholds (See Table “Standard G6PD by SD Biosensor – Manufacturer determined thresholds, as per the instructions for use” under Evidence profiles/Research evidence). These were only available for one product, Standard G6PD (SD Biosensor) and yielded results meeting the requirements of sensitivity and specificity stated in the TPP.

Explanation
At a threshold of 30% of normal G6PD activity: A sensitivity of 100% means that all patients who are G6PD deficient (i.e. who have a G6PD activity of less than 30% of normal G6PD activity based on spectrophotometry) will have a semi-quantitative point-of-care test result indicating G6PD deficiency. None of the G6PD deficient patients will have a test result indicating that they are not deficient (i.e. no false negative test result). A specificity of 97.0% means that of all patients who have a G6PD activity above 30% (based on spectrophotometry), 97% will have a semi-quantitative point-of-care test result indicating that they are NOT G6PD deficient. This does include patients with intermediate G6PD activity of whom the test indicates that they are not deficient. On the contrary, 3% of the patients with a normal or intermediate G6PD activity will have a test result indicating that they are deficient (false positive test result).

If a test with these characteristics would be applied in a group of 10 000 patients in which 5% of patients are G6PD deficient (<30% activity), then this would result in a total of 785 positive test results and thus 785 patients for whom a primaquine regimen of 0.75 mg/kg once weekly for 8 weeks would be suggested, while 285 of them actually have a normal or intermediate G6PD activity. This would also result in 9215 negative test results, who will all have indeed normal or intermediate G6PD activity.

At a threshold of 70% normal G6PD activity: A sensitivity of 91.4% means that of all patients who are G6PD deficient or have intermediate activity (i.e. who have a G6PD activity of less than 70% based on spectrophotometry), 91.4% will have a semi-quantitative near-patient test result indicating G6PD activity below 70%. Thus, 8.6% of the G6PD deficient patients or intermediate will have a test result indicating that they are not deficient (false negative test result). A specificity of 93.7% means that of all patients who have a G6PD activity above 70% (based on spectrophotometry), 93.7% will have a semi-quantitative near-patient test result indicating that their G6PD activity is above 70%. On the contrary, 6.3% of the patients with a normal G6PD activity will have a test result indicating that they are deficient or intermediate (false positive test result).

If a test with these characteristics would be applied in a group of 10 000 patients in which 10% of patients have a G6PD activity below 70%, then this would result in a total of 1055 positive test results and thus 1055 patients who would not be eligible to receive tafenoquine or high dose primaquine (1 mg/kg/day for 7 days), while 598 of them actually have a normal G6PD activity. This would also result in 8945 negative test results and thus 8945 patients who are eligible to receive tafenoquine or high dose primaquine (1 mg/kg/day for 7 days), of whom 43 will have intermediate or even deficient G6PD activity.

Practical info
Although it may be important to know whether a female has an intermediate G6PD activity, programmatically it may not be feasible to tailor the treatments for women differently than for men. Practical guidance should cover all aspects of safe implementation of a new diagnostic test e.g. implementation plan, clear national guidelines, quality assurance/prequalification of tests, training of users, quality assurance of testing, and the type of health care facilities where these tests should be deployed (different levels of health facilities).

There will be additional considerations that are specific for semi-quantitative G6PD tests. Programmes may use Standard G6PD test used with STANDARD G6PD Analyzer (by SD Biosensor, Inc) to identify patients with ≥ 70% of normal G6PD activity to safely administer tafenoquine or high dose primaquine anti-relapse treatment (1mg/kg/day for 7 days) and to identify patients with <30% of normal G6PD activity to administer the low dose weekly primaquine regiment (0.75 mg/kg/week for 8 weeks). Although the systematic review did not explicitly assess the certainty of the evidence to identify patients with ≥ 30% and <70% of normal G6PD activity, this semiquantitative test can be used to safely identify --this group of patients and administer lower dose primaquine anti-relapse treatment (0.5 mg/kg/day for 14 days or 0.5 mg/kg/day for 7 days).

The semi-quantitative tests do provide information about whether a patient has a G6PD activity that falls in the range between 30 and 70. However, the sensitivity to determine whether a woman has a G6PD activity below 70% when an activity below 30% has already been ruled out, is only 52.9% (95% CI 28.8% to 75.7%); so many women with an intermediate activity will be considered “normal” by the semi-quantitative test. On the other hand, the summary specificity is 94.7% (95% CI 86.3% to 98.0%), indicating that only 5.3% of women with “normal” G6PD activity will be told that they have intermediate activity.

Depending on endemicity or regional differences, people may value the importance of the risk of relapses differently, the costs of testing (including training and equipment) may vary, and whether the test would indeed influence management and treatment decisions may vary. However, semi-quantitative tests for G6PD deficiency enable optimal malaria treatment while at the same time limiting the risk of haemolysis.

Conclusion
We are highly confident in the estimates of sensitivity and specificity for the threshold of 30%; the true sensitivity and specificity are very likely to be close to the estimates of the sensitivity and specificity mentioned here. See Table “Certainty of the evidence table for STANDARD G6PD Test (SD BIOSENSOR, Inc): threshold 30%” under Evidence profiles/Research evidence.

We are moderately confident in the estimates of sensitivity and specificity for a threshold of 70%; the true sensitivity and specificity are likely to be close to the estimates of the sensitivity and specificity mentioned here, but there is a possibility that especially the sensitivity may be substantially different. This is because the estimates of sensitivity had very wide confidence intervals. See Table “Certainty of the evidence table for STANDARD G6PD Test a (SD BIOSENSOR, Inc): threshold 70%” under Evidence profiles/Research evidence.

We are less certain for other estimates of the evidence of test’s effects and the evidence of effects on clinical management (the links between test results and management decisions), because of the absence of observational no studies or availability of modelling studies only.

(i) In this document a specific diagnostic test, STANDARD G6PD, is mentioned as the commercial name assigned by the company SD Biosensor Inc to a test and the analyzer they produce. This test is considered by WHO a G6PD semi-quantitative test. WHO considers the quantitative spectrophotometric assay as the reference (gold standard) test for G6PD activity *[266]*.

Values and preferences

The contextual factors review included 10 studies on values, mainly for quantitative tests. These studies found that there is a variation between health care workers in how they value the outcomes of interest (true and false positive and negative test results plus the subsequent consequences thereof). Still, governments ask for evidence regarding the value of G6PD testing as a way to prevent haemolysis cases while still being able to treat patients for malaria. For most outcomes, no or little variation in values may be excepted for the false positives, considered G6PD deficient. These will receive an effective weekly primaquine treatment with lower protection from relapses due to lower adherence to treatment compared to the daily regimens of primaquine, while they do not need protection against risk of haemolysis induced by primaquine.

Resources

The cost of the required analyzer device is probably the biggest differentiator between the cost for implementation of qualitative and semi-quantitative costs, since the semi quantitative has a big initial cost (but cost may be decreased if more persons are being tested). Furthermore, resources are needed to train personnel, and to maintain the test kits and readers; and there may be other logistic costs. On this basis, the resources required were considered to be moderate. There would also be savings because of fewer relapses and fewer haemolysis and less hospitalizations. Although limited research has been conducted to determine the cost-effectiveness of near-patient semi-quantitative

	G6PD tests, the available results suggest a high likelihood that these tests could be cost-effective and potentially cost saving depending on the setting.
Equity	In situations with very few *P. vivax* or *P. ovale* malaria cases and hard to reach areas with limited resources, the initial cost of the device may be a barrier and may thus decrease equity. But insofar as testing allows overall more effective treatment for malaria, there will tend to be an overall positive effect on equity, given the skewed distribution of malaria (towards low income population groups) worldwide.
Acceptability	The semi-quantitative tests are considered probably acceptable, according to the findings of the scoping review of contextual factors (Barker et al., unpublished reference).
Feasibility	The implementation of semi-quantitative test is considered to be feasible to implement, although there may be settings where there may be barriers to the use of qualitative tests.

Justification

Although the certainty about the sensitivity at the threshold of 70% is moderate due to imprecision, the numbers of false negatives will mainly include patients whose G6PD activity will be slightly less than 70% (and thus patients with a lower risk of haemolysis than those with a G6PD activity of <30%).

Without testing, the G6PD activity status of malaria patients remains unknown, which may lead to totally refraining from anti-relapse treatment at all. As this would lead to suboptimal treatment for malaria on a large scale (and many more relapses), we consider that there is a large positive effect of semi-quantitative testing for G6PD deficiency (defined as <30% G6PD activity) on choosing the right treatment to be large. In addition, it will be particularly important to determine the G6PD status for females since those with intermediate activity of G6PD (30–70%) will have a higher risk of haemolysis than patients who have a G6PD activity above 70%.

Research needs

The scoping review on contextual factors showed that there is an evidence need around contextual factors, especially for the evidence of the effects of G6PD testing implementation on haemolysis, effects and clinical management (the links between test results and management decisions) and equity and human rights issues.

More evidence may be needed to provide a higher certainty of the diagnostic accuracy evidence of all near-patient semi-quantitative G6PD tests available for all three categories of G6PD activity: deficiency, intermediate, and normal G6PD activity.

More evidence may be need on the cost-effectiveness and potential cost-saving achievable through the use of semi-quantitative versus qualitative G6PD tests.

5.2.1.7 Anti-relapse treatment of P. vivax and P. ovale

Conditional recommendation for , Low certainty evidence

Tafenoquine as anti-relapse therapy (2024)

Tafenoquine is recommended as an alternative to primaquine (3.5 mg/kg total dose) for preventing relapses of *P. vivax* in patients ≥ 2years of age, who have ≥ 70% G6PD activity and who receive chloroquine treatment.

- *These recommendations pertain only to South America.*
- *Quantitative or semi-quantitative determination of G6PD activity must be done before tafenoquine administration.*
- *Tafenoquine is not recommended in pregnant and lactating women.*
- *Tafenoquine is not recommended in patients receiving artemisinin-based combination therapies for the treatment of P. vivax.*
- *Controlled deployment and /or further research is encouraged outside of South America, to generate evidence of the efficacy and safety of tafenoquine compared to primaquine as an anti-relapse treatment.*
- *No data is available comparing tafenoquine with primaquine given at a total dose of 7.0 mg/kg.*

Practical info

- Quantitative or semi-quantitative G6PD testing MUST be done prior to tafenoquine administration.
- Dose:
  - Adults, adolescents and children weighing more than 35 kg: the administration of a single 300 mg dose (two 150 mg tafenoquine tablets) is recommended on Day 1 or Day 2 of the 3-day course of chloroquine
  - Children ≥ 2 years of age and weighing >10 kg to ≤ 35 kg:

Dispersible tablet dose recommendations for children (>10 kg to ≤ 35 kg)

Body weight (kg)	Total dose	Number of tablets
> 10 to ≤ 20	100 mg	Two 50 mg dispersible tablets
> 20 to ≤ 35	200 mg	Four 50mg dispersible tablets

- Tafenoquine is NOT recommended for pregnant or lactating women.

Evidence to decision

Benefits and harms

Tafenoquine is probably as effective as primaquine (3.5 mg/kg total dose) for *P. vivax* relapse prevention up to 6 months. The 3.5 mg/kg total primaquine dose is given as either 0.25 mg/kg/day for 14 days or 0.5 mg/kg/day for 7 days; the two regimens have similar efficacy and safety. Tafenoquine + chloroquine probably has little or no difference in adverse events (any type) compared to primaquine (3.5 mg/kg total dose) + chloroquine. Although there was a slightly greater drop in haemoglobin with tafenoquine + chloroquine compared to primaquine + chloroquine, the overall balance between benefit and harm does not seem to favor either tafenoquine or primaquine *[267]*.

The recommendation is only presently applicable to South America as the evidence on efficacy and safety is mainly from studies in South America (contributing to over 75% of the sample size data). Also, tafenoquine comparison was only with primaquine given at the total dose of 3.5 mg/kg. No data is available on the comparison of tafenoquine with the currently recommended higher primaquine total dose of 7.0 mg/kg.

Limited data from South-East Asia indicates that chloroquine with primaquine (3.5 mg/kg) performed better in preventing relapses than chloroquine with tafenoquine *[268]*. This finding seems to imply that the response to tafenoquine, as compared to primaquine, may vary in different regions. More evidence on the safety and efficacy of tafenoquine is required for other regions, also in comparison to the currently recommended higher dose of primaquine (7.0 mg/kg total dose).

Tafenoquine should not be deployed without accurate G6PD quantitative or semi-quantitative testing and should only be used if patients have ≥70% G6PD activity.

Certainty of the evidence	Low Though the certainty of the evidence was moderate for the efficacy outcome of parasitaemia during the six months of follow-up, the overall certainty of the evidence was rated low because of the serious imprecision regarding serious adverse events.
Values and preferences	The values were considered relatively homogeneous, i.e., there was probably no important uncertainty or variability. It is difficult to assess how different populations would value benefits versus a potentially serious adverse reactions, but considerations of preferences and values do not favour either tafenoquine or primaquine.
Resources	Presently, the cost of tafenoquine and a full course of primaquine are approximately the same. However, the deployment of tafenoquine would entail greater resource requirements, resulting in larger costs. This higher cost is due to the requirement for quantitative or semi-quantitative determination of G6PD activity to safely administer tafenoquine as against the requirement of a qualitative test to safely administer primaquine at 3.5 mg/kg. The cost to quantitatively or semi-quantitatively measure G6PD activity (including the analyser and the test strips) is high. However, the available economic evaluations showed that tafenoquine prescribed after semi-quantitative G6PD testing is highly likely to be cost-effective in Brazil, considering a willingness to pay threshold of US$ 7800 adopted by the Brazilian National Commission for the Incorporation of Technologies (CONITEC), considered the maximum value that an intervention should cost per DALY averted to be considered cost-effective *[269]*.
Equity	There is a theoretical risk for reduced health equity since all patients with G6PD activity below 70% of normal would be excluded from receiving tafenoquine. However, this risk is mitigated by the availability of primaquine, which remains an option for anti-relapse treatment for these patients. It is expected that all countries that deploying tafenoquine will also need to continue to deploy primaquine as an anti-relapse medicine.
Feasibility	The introduction of tafenoquine as an alternative drug is probably feasible. The evidence from a study in Brazil showed that 99·7% (95% CI 99·4 to 99·8) of patients with P. vivax monoinfection aged ≥16 years were treated or not treated with tafenoquine in accordance with G6PD activity assessed by point-of-care quantitative tests *[270]*. This result was consistent across all health facilities. Across all patients who received tafenoquine, 97·7% (95%CI 97·0 to 98·2; 2623/2685) were treated according to the treatment algorithm. Chloroquine was administered to 99·9% (2682/2685) of patients with tafenoquine. Forty-nine patients presented with mixed infection, and all five who received tafenoquine were G6PD normal. The study showed that mistakes in administration were more likely to be associated with prescribing errors rather than errors in the quantitative G6PD test.

Research needs

Research is needed to generate further evidence on the efficacy and safety of tafenoquine in other regions of the world than South America.

Strong recommendation for , Moderate certainty evidence

Primaquine as anti-relapse therapy (2024)

To prevent relapse, children and adult (except pregnant women, infants aged < 1 months and women breastfeeding infants aged < 1 months, and people with G6PD deficiency), primaquine should be given at a high total dose (7 mg/kg) at 0.5 mg/kg/day for 14 days or 1 mg/kg/day for 7 days for prevention of relapses in patients with uncomplicated *P. vivax* or *P. ovale* malaria.

- *The primaquine high dose (7 mg/kg) should be provided at 1 mg/kg/day for 7 days only to patients with ≥70% G6PD activity.*
- *National decisions regarding the two high-dose (7 mg/kg) primaquine regimens given over 7 or 14 days will be affected by the availability of G6PD semi-quantitative testing and capacity for supervised therapy.*
- *Evidence for the magnitude of benefit may vary geographically. Whether a high dose of primaquine 7 mg/kg is given in 14 days or 7 days, the absolute benefit of using the high primaquine total dose will vary according to the risk of recurrence in the population. The benefits are higher in Africa, South-East Asia and Oceania. However, in areas on the Indian subcontinent and in the Americas, where the absolute benefit of a total high dose of 7 mg/kg might be only marginally greater than that of 3.5 mg/kg, primaquine at a low 3.5 mg/kg total dose might be used.*
- *It should be emphasized that determination of G6PD status using appropriate test is needed to guide the safe administration of primaquine (see section 5.2.1.6 on G6PD testing).*

Practical info

1. It is appreciated that national decisions regarding the choice of high dose PQ regimens (7.0 mg/kg total dose given at 0.5 mg/kg/day for 14 days or 7.0 mg/kg total dose given as 1 mg/kg/day for 7 days) as against the low dose PQ regimen (3.5 mg/kg total dose given at 0.5 mg/kg/day for 7 days) will depend on the risk-benefit analysis depending on local *P. vivax* relapse rate, the availability of different types of G6PD tests, and capacity for supervised therapy. Our recommendations were based on supervised or semi-supervised PQ administration. Effectiveness might be reduced in unsupervised scenarios.
2. Evidence for magnitude of benefit may vary geographically. Primaquine at 3.5 mg/kg total dose for patients with uncomplicated *P. vivax* malaria might be used in South Asia and the Americas where the absolute benefit of high dose might be small depending on the absolute risk of recurrence following low dose 8-aminoquinolines therapy.
3. It should be emphasized that G6PD testing is needed prior to primaquine administration.
4. Primaquine can be given except for pregnant women, infants < 1 months of age and women breastfeeding infants < 1 months of age. Secretion of primaquine in breast milk is negligible *[229]*.

Evidence to decision

Benefits and harms	The systematic review showed moderate benefit and small undesirable effects, probably favoring primaquine at a high total dose of 7.0 mg/kg over primaquine at a low total dose of 3.5 mg/kg *[308]*. There may be a moderate to large reduction in *P. vivax* recurrences with 7 mg/kg total dose primaquine compared with 3.5 mg/kg total dose primaquine (moderate certainty of evidence across all patient groups, moderate certainty of evidence in children <5 years). However, the evidence is uncertain about the effect of 7 mg/kg total dose primaquine compared with 3.5 mg/kg total dose primaquine in the Americas and South Asia (very low certainty of evidence) *[309]*. Adverse events did not increase in patients treated with 14 days of primaquine at 0.5 mg/kg/day (7 mg/kg) compared with 7 days of primaquine at 0.5 mg/kg/day (3.5 mg/kg). Clinically relevant haemolysis is rare. However, the daily dose rather than the total dose of primaquine has a significant effect on the risk of clinically relevant haemolysis and gastrointestinal intolerance. In patients with intermediate G6PD activity (30-<70%) there may be a significant increase in the risk of clinically relevant haemolysis (>25% fall to Hb <7 g/dL) in those treated with 1 mg/kg/day primaquine for 7 days compared with 0.25 mg/kg/day primaquine for 14 days (low certainty of evidence), while in patients with 70% G6PD there may be a very small to no increase in the risk of clinically relevant haemolysis (very low certainty of evidence). Gastrointestinal intolerance may be increased at days 5-7 in people receiving 1mg/kg/day primaquine for 7 days compared with 0.25 mg/kg/day primaquine for 14 days (very low certainty of evidence).

Certainty of the evidence

Moderate

Although the evidence summary for the different outcomes showed variable certainty of evidence (efficacy: high to very low, undesirable outcomes: low to very low), the panel decided to give more weight to the certainty of evidence for moderate efficacy outcome.

Values and preferences

There is possibly important uncertainty or variability on how the outcomes are valued by patients and stakeholders.

Whether a high dose of primaquine 7 mg/kg is given over 14 days or 7 days, the absolute benefit of increasing the total dose will vary according to the risk of recurrence in a population. There may be a variation in preferences in different settings depending on how national programmes value the impact of shorter regimens which may lead to improved adherence and effectiveness *[310]*. The risk of clinically relevant haemolysis will vary according to the prevalence of G6PD deficiency in the population and the ability to test for G6PD deficiency.

It was noted that the evidence for this recommendation was based on clinical trial settings with close supervision of administration of the treatment options. This might have an impact on how outcomes will be valued in different settings.

Resources

There is low certainty on whether there will be negligible costs and savings because costs of required resources may vary geographically.

There will be an increased cost for higher primaquine dosing given that the dose is doubled. Prescriber training and end-user information will be needed to improve prescribing, compliance and adverse event awareness. The primaquine high dose regimen of 7 mg/kg total dose given as 1 mg/kg/day for 7 days requires a semi-quantitative G6PD assay to target treatment to patients who have ≥70% G6PD activity and the cost for the test and analyzer needs to be added to the costing for resources. The need for follow-up and management of adverse events should also be considered. On the other hand, a reduction in the risk of recurrence will lead to reduced direct and indirect costs on the healthcare system and reduced household costs.

Evidence coming from cost evaluation studies showed that high dose primaquine is probably favored over low dose primaquine. A global cost-effectiveness analysis, undertaken prior to the updated evidence for the efficacy of high dose compared with low dose primaquine, assessed the cost of implementing high dose primaquine over 14 days after assessing G6PD activity using a point of care test to identify individuals with <30% and >30% G6PD activity. The study concluded that a substantial global economic burden of vivax malaria could be reduced through investment in safe and effective radical cure achieved by routine screening for G6PD deficiency and supervised treatment *[274]*.

Equity

The panel assessed that the proposed intervention of high-dose primaquine probably will have no impact on equity, although it may vary across different populations. Enhancing treatment efficacy and reducing relapses will probably increase equity, as the burden of relapses is disproportionately higher in populations with reduced resources. There might be a possibility of reduced equity if poorer populations have limited access to G6PD testing prior to therapy.

Heterozygous female patients with intermediate G6PD activity (30-<70%) may have an increased risk of adverse events from a high dose 7-day primaquine regimen *[312]*.

Acceptability

Acceptability could vary because of the variability in the effectiveness and risk for adverse events in different populations. In populations using a 14-day primaquine dose of 0.25 mg/kg/day, the improved efficacy associated with increasing the dose of primaquine from 0.25 to 0.5 mg/kg/day may support acceptability from the perspective of patients and their carers. However, the increased pill burden also needs to be considered. From a healthcare provider perspective, the increased importance of G6PD testing and local pharmacovigilance systems may decrease acceptability.

In populations using a 7-day primaquine dose of 0.5 mg/kg/day, the improved efficacy of a 14-day dose of 0.5 mg/kg/day may support acceptability from the perspective of patients and healthcare providers. However, reduced acceptability of a longer duration will likely be important in some populations.

In populations using a 7-day primaquine regimen of 1 mg/kg/day, the improved efficacy in most populations and short duration of treatment will increase the acceptability for most users. However, this may be countered by the increased pill burden associated with a higher daily dose. Reduced gastrointestinal tolerability of the 1 mg/kg/day dose may reduce acceptability, although this may be mitigated by intake with food. The local availability of G6PD semiquantitative testing and local pharmacovigilance systems may impact the acceptability for some implementers to change policy, especially given the potential for increased haemolysis in individuals with 30–<70% G6PD activity with a 7-day 1 mg/kg/day primaquine regimen. Community engagement and education of healthcare workers, patients and their carers may enhance acceptability and ensure awareness of early indicators of adverse effects.

The decision regarding total dose may impact the duration of therapy or the decision to implement specific types of the G6PD testing.

Feasibility The administration of a high dose instead of low dose primaquine is probably feasible.

Implementation of high total dose primaquine over 14 days or 7 days will be feasible in settings where G6PD testing is available and/or pharmacovigilance is in place.

There were concerns expressed by the panel about the availability and operational challenges of the G6PD test at present.

Research needs

Studies on how the difference in dosing and duration can affect compliance are needed with real-life evaluation of impact on recurrence rates. Pharmacovigilance (including at the community level) and more studies on acceptability of G6PD testing prior to administration of different regimens of primaquine should be conducted.

Conditional recommendation for , Very low certainty evidence

Preventing relapse in people with G6PD deficiency (2015)

In people with G6PD deficiency, primaquine base at 0.75 mg/kg bw once a week for 8 weeks can be given to prevent relapse, with close medical supervision for potential primaquine-induced haemolysis.

Practical info

- In patients known to be G6PD deficient, primaquine may be considered at a dose of 0.75 mg base/kg bw once a week for 8 weeks. The decision to give or withhold primaquine should depend on the possibility of giving the treatment under close medical supervision, with ready access to health facilities with blood transfusion services.
- Some heterozygote females who test as normal or not deficient in qualitative G6PD screening tests have intermediate G6PD activity and can still haemolyse substantially. Intermediate deficiency (30–70% of normal) and normal enzyme activity (> 70% of normal) can be differentiated only with a quantitative or semiquantitative G6PD test. In the absence of quantitative testing, all females should be considered as potentially having intermediate G6PD activity and given the 14-day regimen of primaquine, with counselling on how to recognize symptoms and signs of haemolytic anaemia. They should be advised to stop primaquine and be told where to seek care should these signs develop.
- If G6PD testing is not available, a decision to prescribe or withhold primaquine should be based on the balance of the probability and benefits of preventing relapse against the risks of primaquine-induced haemolytic anaemia. This depends on the population prevalence of G6PD deficiency, the severity of the prevalent genotypes and on the capacity of health services to identify and manage primaquine-induced haemolytic reactions.

Evidence to decision

Benefits and harms	Desirable effects: • There are no comparative trials of the efficacy or safety of primaquine in people with G6PD deficiency. Undesirable effects: • Primaquine is known to cause haemolysis in people with G6PD deficiency. • Of the 15 trials included in the systematic review, 12 explicitly excluded people with G6PD deficiency; in three trials, it was unclear whether participants were tested for G6PD deficiency or excluded. None of the trials reported serious or treatment-limiting adverse events.
Certainty of the evidence	Very low Overall certainty of evidence for all critical outcomes: very low.

Justification

GRADE

In a systematic review of primaquine for radical cure of *P. vivax* malaria *[276]*, 14 days of primaquine was compared with placebo or no treatment in 10 trials, and 14 days was compared with 7 days in one trial. The trials were conducted in Colombia, Ethiopia, India, Pakistan and Thailand between 1992 and 2006.

In comparison with placebo or no primaquine:

14 days of primaquine (0.25 mg/kg bw per day) reduced relapses during 15 months of follow-up by about 40% (RR, 0.60; 95% CI, 0.48–0.75, 10 trials, 1740 participants, high-quality evidence).

In comparison with 7 days of primaquine:

14 days of primaquine (0.25 mg/kg bw per day) reduced relapses during 6 months of follow-up by over 50% (RR, 0.45; 95% CI, 0.25–0.81, one trial, 126 participants, low-quality evidence).

No direct comparison has been made of higher doses (0.5 mg/kg bw for 14 days) with the standard regimen (0.25 mg/kg bw for 14 days).

Twelve of the 15 trials included in the review explicitly excluded people with G6PD deficiency; the remaining three did not report on this aspect. No serious adverse events were reported.

Other considerations

In the absence of evidence to recommend alternatives, the guideline development group considers 0.75 mg/kg bw primaquine given once weekly for 8 weeks to be the safest regimen for people with G6PD deficiency.

Primaquine and glucose-6-phosphate dehydrogenase deficiency

Any person (male or female) with red cell G6PD activity < 30% of the normal mean has G6PD deficiency and will experience haemolysis after primaquine. Heterozygote females with higher mean red cell activities may still show substantial haemolysis. G6PD deficiency is an inherited sex-linked genetic disorder, which is associated with some protection against *P. falciparum* and *P. vivax* malaria but increased susceptibility to oxidant haemolysis. The prevalence of G6PD deficiency varies, but in tropical areas it is typically 3–35%; high frequencies are found only in areas where malaria is or has been endemic. There are many (> 180) different G6PD deficiency genetic variants; nearly all of which make the red cells susceptible to oxidant haemolysis, but the severity of haemolysis may vary. Primaquine generates reactive intermediate metabolites that are oxidant and cause variable haemolysis in G6PD-deficient individuals. It also causes methemoglobinaemia. The severity of haemolytic anaemia depends on the dose of primaquine and on the variant of the G6PD enzyme. Fortunately, primaquine is eliminated rapidly so haemolysis is self-limiting once the drug is stopped. In the absence of exposure to primaquine or another oxidant agent, G6PD deficiency rarely causes clinical manifestations so, many patients are unaware of their G6PD status.